ZUSE Automat Agent: Empirical Law Discovery in Elementary Cellular Automata

Abstract

We present ZUSE Automat Agent (ZUSE), a deterministic, policy-driven discovery loop for elementary cellular automata (ECA) that builds an empirical atlas of cycle laws, world families, and basin fragility without a language model in the discovery loop. The pipeline combines a fixed seven-law evaluator, a dedup-gated observer stack, and persistent multi-seed world records, and is applied to a 20-world sample spanning ECA rules, Conway's Game of Life patterns, and synthetic controls.

The seven laws — *velocidad_constante*, *periodicidad*, *densidad_estable*, *tipo_unico*, *complejidad_alta*, *frontera_temporal*, and *temporal_scale_stability* — are calibrated empirically (*frontera_temporal* upper bound 0.4352; *temporal_scale_stability* threshold 19.03, decision-tree accuracy 0.908). Laws are separated into two groups: structure-observer laws that depend on the full observer stack, and frame-metric laws that depend only on aggregate frame statistics.

Key results: (1) *frontera_temporal* is not intrinsically rare — it activates in 38 of 256 ECA rules under at least two of three seeds, and in 17 of 256 under all three; (2) the 20-world atlas reveals seven dynamic categories (*frontera-rich-estable*, *periodicidad-global*, *oscilador-local*, *multiregimen-productivo*, *multiregimen-escala-dependiente*, *noise-bounded*, *sin-evidencia-multiregimen*), finer than Wolfram's four-class taxonomy; (3) one-bit IC fragility spans from perfectly stable basins (`rule_208/209`, `f_total = 0.000`) to exact fixture disruption (`life_blinker`, `f_total = 1.000`), with `rule_108` as the ECA outlier (`f_total = 0.992`, `f_gap = 0.945`). These cases separate three mechanistically distinct regimes: productive basin switching, noise-boundary crossing, and quiescent-background activation; (4) an exhaustive protocol over 128 quiescent ECA rules and 502 non-zero IC words per rule confirms `rule_108` as the unique ECA rule producing stationary local period-2 oscillators; (5) designed periodic ICs activate production `periodicidad` in 207 of 256 ECA rules, showing that the law is IC-family sensitive rather than inaccessible; and (6) a controlled single-bit experiment demonstrates that ECA frames are translation-invariant while the observer/dedup pipeline is not, separating physical law from measurement artifact; and (7) three progressive background-oscillator sweeps reveal that the local oscillator landscape is strongly background-conditioned. Under quiescent zero background, only `rule_108` (stationary, T=2) and eight rules (moving, T=2, speed 1 cell/step) produce local oscillators. Under non-zero periodic backgrounds of template length 1, 2, and 4 (1,927,680 runs), the landscape expands to 30 stationary and 36 moving rules, introducing period-4 oscillators and speed-0.5 gliders. Under 30 primitive length-8 binary backgrounds (3,855,360 runs), 23 further rule/type pairs appear, the observed period extends to T=15, and speed 2/3 cell/step is observed for the first time. Phase sensitivity is detected in all 10 sampled rules from the length-8 sweep; a strict IC/background co-translation test confirms exact physical equivariance in 80/80 runs after correcting a cyclic-boundary artifact in `linear_shape`. The `T=15` family is confined to the reflection-symmetric, black/white-conjugate pair `rule_73/rule_109`, locks at five times the background temporal period, and persists through step 900 in all 20 minimal rule/background representatives. Sampling the localized defect once per background period reveals a minimal five-state cycle under `F^3` in 20/20 representatives, establishing the computational mechanism of the measured 5:1 locking ratio. The induced defect rule `delta_f(b,d)=f(b XOR d) XOR f(b)` further yields an exact analytical conjugation identity between `rule_73` and `rule_109`; exhaustive checks reject a fixed sparse truth-table support as the universal source of the cycle. The remaining symbolic state reduces to 13 finite defect-cycle families and a compact length-8 descriptor `(rule, subpatterns_len4, IC/background alignment)`, validated by 140/140 co-translated rotations. A targeted external length-9/10 test over 66 `T_bg=3` backgrounds finds 90 additional T=15 detections across 8 new backgrounds, showing that the mechanism is not confined to primitive length 8. Subsequent transition-table and pre-burn-in analyses show that the post-burn-in macro-operator refines the 13 visual families, but that the entry phase before burn-in does not admit a compact local predictor in the tested descriptors: all 20 representatives enter the stable five-cycle by `t=12`, yet exact pre-burn-in predictors mostly collapse to singleton case identifiers. A targeted causal-cone test resolves this negative result constructively: a strict 25-cell cone simulated for 12 steps recovers the full post-burn-in `defect_state0` in 20/20 representatives after phase projection, a 69.1x compression over the full 256-by-81 simulation. Auditing that cone shows no sparse subtable shortcut: all 25 initial cone inputs and all eight ordinary ECA entries remain active, leaving Boolean simplification of the dense local circuit as the remaining symbolic target. A reduced ordered BDD audit confirms that the active-output functions still depend on all 25 inputs under the natural cone order, ruling out Boolean input elimination. A subsequent order-sensitivity and targeted SIFT audit finds no useful BDD-size shortcut: simple reversal improves active nodes by only 0.5% globally, and a 580-order SIFT pass on the most favorable representative improves 16,061 active nodes to 16,056, far above the 10,000-node compression gate. ANF analysis of the same 25-by-12 causal cone reveals a spatial algebraic-complexity gradient: active-output degree obeys `degree = 24 - d + epsilon`, with `epsilon in {0,1}` and zero exceptions over 174 active outputs, while monomial count decays approximately as `2^{-d}` with distance `d` from the defect center (`R^2 = 0.998197`). A follow-up epsilon-residual audit finds no compact predictor from static rule, position, local background, family, or cycle-phase features: the best single feature reaches 64.89% leave-one-representative-out accuracy, while a depth-3 decision tree reaches 73.05% training accuracy but only 55.65% leave-one-representative-out accuracy. Recomputing the ANF profile layer-by-layer over the full 12-step cone reverses that negative result: `degree_growth_slope` predicts the epsilon bit with 94.90% leave-one-representative-out accuracy and zero mismatches against the final Fase 44 ANF degrees. A horizon audit then shows that this is a full-profile effect: partial horizons up to `K=11` remain below 80% LORO accuracy, while `K=12` jumps to 94.90%. The ANF gradient also generalizes across external length-9/10 `T=15` witnesses with 0/63 band exceptions and `R^2 = 0.998263`. Follow-up baselines outside the original `T=15` setting show that the gradient

is not caused by period, cone size, or active-support width alone: compact T=2 oscillators lack enough active support, an external `rule_109`/`T=10` witness reproduces the T=15 slope within 0.13%, and tested external families (`rule_54`, `rule_94`, `rule_133`) are flat at their natural periods. A targeted family robustness test first isolates the known non-T15 witness, but a full catalog census then finds additional `rule_109` witnesses: two `T=12` cases are strong at their natural period, and two `T=8` cases are acceptable at the common 12-step horizon. No `rule_73` or external-family case becomes a natural-period or acceptable-horizon witness in the census. The current evidence therefore identifies the gradient as mechanism-dependent and specifically concentrated in `rule_109` within the tested catalog, without claiming universality over untested ECA families.

Every result is reproducible from deterministic scripts with no stochastic components in the discovery loop.

1. Introduction

Elementary cellular automata are among the simplest systems known to exhibit complex behavior. A radius-1, binary, one-dimensional CA is fully specified by a single integer from 0 to 255, yet even within this minimal space, Wolfram's empirical taxonomy finds four qualitatively distinct dynamic classes: uniform, periodic, locally chaotic, and complex. Cook's proof that Rule 110 supports universal computation establishes that complexity in ECA is not merely visual: it has computational consequences.

Two questions remain largely open after Wolfram's program. First, the taxonomy is qualitative and coarse: all complex rules fall into Class 4 regardless of their intra-class differences in structure type, periodicity, fragility, or scale behavior. Second, the boundary between the dynamics of the rule and the properties of the measurement instrument is rarely made explicit. When an observer reports that a run contains gliders, it conflates the CA physics with the heuristic that defined the glider label.

ZUSE Automat Agent addresses both questions through a deterministic, policy-driven discovery loop. The agent runs ECA worlds, applies a fixed stack of heuristic observers, evaluates seven binary cycle laws, and stores multi-seed evidence in persistent world records. No language model participates in the discovery loop: law proposals, world selection, and evidence evaluation are all deterministic. Language-model assistance is restricted to post-run interpretation and documentation.

The result is an empirical atlas of 20 worlds with seven operational categories, measured fragility along two axes (`f_total` and `f_core`), and explicit characterization of two observer artifacts. The atlas is not a new taxonomy of Wolfram's classes; it is a finer-grained, evidence-based map of a 20-world sample that separates cycle-level laws, world-level regimes, and pipeline behavior.

1.1 Contributions

We make the following contributions:

1. **ZUSE Automat Agent** — A deterministic, policy-driven discovery loop for ECA that accumulates multi-seed law evidence across worlds without symbolic regression or LLM guidance in the loop. The agent combines persistent world-record history, a dedup-gated observer stack, and a seven-law evaluator into a single reproducible pipeline.

2. **A seven-category empirical atlas of 20 worlds** — We classify 20 worlds spanning ECA rules, Conway's Game of Life patterns, and synthetic controls into seven operational categories (*frontera-rich-estable*, *periodicidad-global*, *oscilador-local*, *multiregimen-productivo*, *multiregimen-escala-dependiente*, *noise-bounded*, *sin-evidencia-multiregimen*) using law coverage, signature diversity, and fragility. The atlas extends Wolfram's four-class taxonomy by capturing intra-class structure, scale-dependent silencing, and negative-control regimes.

3. **A two-dimensional fragility framework** — We measure `f_total` and `f_core` separately, defining `f_gap = f_total - f_core` as a quantitative measure of secondary-law churn. Four distinct mechanisms are identified: stable basin (`rule_208/209`, `f_total = 0.000`), productive basin switching (`rule_137`, `f_gap = 0.318`), noise-boundary fragility (`rule_54`, `f_core = 0.677`), and quiescent-background activation (`rule_108`, `f_gap = 0.945`).

4. **`rule_108` as the unique stationary local-period-2 ECA oscillator** — Under an exhaustive protocol (128 quiescent ECA rules, 502 non-zero IC words per rule, span <= 32, period <= 16), `rule_108` is the only ECA rule that produces stationary local period-2 oscillators. The motif `#.# <-> ###` follows algebraically from `f(0,1,0) = f(1,0,1) = 1` and `f(1,1,1) = 0`, and the rule's left-right symmetry (`f(l,c,r) = f(r,c,l)`) explains why the oscillator does not drift.

5. **A measured separation between ECA dynamics and observer artifacts** — `rule_54` single-bit-IC frames are provably translation-invariant (confirmed by frame identity after shift normalization), while observer dedup counts range from 15 to 24 across IC positions. This non-equivariance is characterized as a pipeline property: absolute structure counts depend on IC context, but law signatures remain stable.

2. Related Work

ZUSE sits at the intersection of three bodies of prior work: empirical ECA taxonomy, automated law discovery, and LLM-based scientific agents. It is related to each tradition but positioned differently from all three: it extends ECA taxonomy inward (intra-class rather than inter-class), it inverts the discovery framing of symbolic regression systems (fixed evaluators rather than generative hypotheses), and it excludes the LLM from the loop rather than centering it.

2.1 ECA taxonomy and complexity

Wolfram's systematic study of elementary cellular automata established the canonical four-class taxonomy: Class 1 (uniform), Class 2 (periodic), Class 3 (chaotic), and Class 4 (complex) [Wolfram2002]. This taxonomy is qualitative and based on visual inspection of space-time diagrams. ZUSE extends it by measuring intra-class structure: two Class-4 rules (`rule_137` and `rule_54`) differ not only in fragility magnitude but in fragility mechanism, a distinction the four-class taxonomy does not capture.

Cook's proof that Rule 110 supports universal computation [Cook2004] established that ECA complexity has computational consequences beyond visual appearance. But computational class is a coarse lens: `rule_110` and `rule_54` are both Class 4, yet ZUSE finds that their fragility mechanisms are qualitatively different — productive basin switching versus noise-boundary crossing. `rule_110` appears in the ZUSE atlas as a `multiregimen-productivo` world with `f_total = 0.323` and a stable frontera signature. The computational proof characterizes what `rule_110` *can* compute; ZUSE characterizes what it *typically does* under random initialization.

2.2 Automated scientific discovery

AI Feynman [Udrescu2020] demonstrated symbolic regression over physical datasets, recovering known equations from data with interpretable structure. The contrast with ZUSE is deliberate: AI Feynman uses neural networks to propose candidate laws from continuous-variable data, while ZUSE applies fixed binary evaluators to discrete CA dynamics and accumulates evidence without a generative component. ZUSE is not a symbolic regression system; it is a policy-driven measurement pipeline whose outputs are law signatures, not formulas.

More broadly, systems such as Eureqa [Schmidt2009] and recent LLM-based discovery agents frame discovery as hypothesis generation followed by verification. ZUSE inverts this framing: laws are fixed a priori, and the discovery consists of finding which worlds satisfy them and under what conditions. This makes every accepted law signature verifiable from deterministic scripts, at the cost of not proposing new laws automatically.

2.3 ZUSE as evidence engine, not LLM scientist

Recent work on LLM-based scientific agents (e.g., The AI Scientist [Lu2024]) demonstrates that language models can propose hypotheses, design experiments, and write papers with minimal human intervention. ZUSE occupies a different position in this space: the language model is explicitly excluded from the discovery loop and restricted to post-run interpretation and documentation.

This separation is a design choice, not a limitation. It means that the atlas findings are fully reproducible from the deterministic loop code, and that language-model involvement can be audited at the documentation layer without contaminating the empirical results. The cost is that ZUSE cannot propose new laws; the benefit is that every accepted law has a transparent, non-generative provenance.

3. System: ZUSE Automat Agent

ZUSE Automat Agent is a deterministic discovery loop over cellular automaton worlds. A world is a simulator plus an initial-condition protocol and a time window. For ECA worlds, the simulator is the standard binary radius-1 update rule with periodic boundary conditions. The agent runs a world, computes frame metrics, extracts candidate structures, evaluates a fixed set of laws, updates world-level history, and chooses the next action through a transparent policy.

3.1 Inputs and outputs

A single agent cycle takes as input:

- **World identifier**: the ECA rule number (`0..255`) or a named synthetic world.
- **IC protocol**: either a random seed for standard runs or a designed IC vector for controlled experiments.
- **Width and steps**: fixed integers, typically `width = 64` and `steps = 24..200`, depending on the world's scale protocol.

The outputs of a single cycle are:

- **Frame metrics**: density mean, entropy mean, temporal transition rate, gzip ratio, and mutual information mean.
- **Analysis status**: `ok` or `ruido_no_analizable` after the noise gate.
- **Structure records**: raw `Estructura` outputs with type labels and span information, plus `dedup_structure_count`.
- **Law signature**: frozenset of accepted law names, or the empty set if the run is noise-gated.

3.2 Loop structure

The loop has five layers:

1. **Simulation.** ECA frames are generated from explicit initial conditions with fixed `width`, `steps`, and `seed` or with designed ICs for controlled experiments. The simulator itself is not learned.
2. **Frame metrics.** Each run is summarized by density, entropy, temporal transition rate, gzip compressibility, and temporal mutual information. These features support both individual laws (`complejidad_alta`, `frontera_temporal`, `temporal_scale_stability`) and later meta-analysis.
3. **Observers and deduplication.** A stack of heuristic observers converts frame histories into `Estructura` records with type labels such as `glider`, `bloque`, and `oscilador`. The raw observer outputs are intentionally kept for audit, while `deduplicate_structures` estimates the number of physical structures. The production noise gate uses `dedup_structure_count > 40`.
4. **Cycle-law evaluation.** Seven laws are evaluated on each analyzable run. The result is a law signature: the set of accepted laws for that cycle. Noise-gated runs skip law evaluation rather than forcing a low-confidence signature.
5. **Policy and memory.** The agent stores a persistent `WorldRecord` per world and chooses the next action after each cycle.

3.3 State: WorldRecord

Each world maintains a `WorldRecord` with the following fields relevant to atlas construction:

field	description
`visit_count`	total cycles run on this world
`scores`	per-cycle scores used by the policy
`noise_count` / `noise_fraction`	count and fraction of noise-gated cycles
`law_signatures`	list of accepted law signatures as frozensets
`unique_law_signature_count`	count of distinct non-empty signatures
`non_empty_signature_visit_count`	visits where at least one law was accepted
`law_signature_diversity`	unique non-empty signatures divided by non-empty visits, reported after at least five non-empty visits
`peak_signature_diversity`	maximum clean diversity observed so far
`has_multiregime_evidence`	monotone boolean, set once peak diversity exceeds `0.5` under low noise
`params_tried`	tested `(steps, width, law_signature)` tuples
`max_ok_steps` / `first_noise_steps`	scale boundary diagnostics

The important design choice is that empty signatures are retained for audit but excluded from diversity. A world is not multi-regime merely because it alternates between laws and silence; multi-regime evidence requires multiple non-empty law signatures.

3.4 Journal

Every cycle result is appended to a JSONL journal (`outputs/experiments_*/journal_*.jsonl`). Each line is a self-contained JSON record with cycle identifier, world identifier, steps, width, frame metrics, analysis status, structure counts, law signature, action taken, and the previous WorldRecord state visible to the policy at decision time. The journal is the primary reproducibility artifact: the atlas in Section 5, the fragility measurements in Section 6, and the case studies in Section 7 are all derived from journal queries and controlled follow-up scripts.

3.5 Policy

The policy selects the next action from four options:

- **Vary seed** (`repeat_vary_seed`): run the same world with a new IC. Used when the world has a new law signature or confirmed multi-regime evidence and the current cycle is productive.
- **Increase scale** (`increase_steps`): raise `steps` to test scale-dependent behavior. Used when the current world produces analyzable signal and has not reached a known noise boundary.
- **Change world** (`change_world`): move to the next world. Used when the current world is noise-bounded, reaches a known noise boundary, has exhausted repeats, or converges to unproductive silence at maximum scale.
- **Stop by exhaustion**: after the requested cycle budget, persist state and journal artifacts.

The policy has no learned parameters. Its thresholds are fixed constants or explicit guards in code: `dedup_structure_count > 40` for the noise gate, signature diversity `> 0.5` for multi-regime evidence, `noise_fraction < 0.20` for clean diversity, and at least five non-empty visits before diversity is reported.

3.6 Non-generative design

The agent is deliberately non-generative inside the loop. No LLM proposes laws, selects worlds, or evaluates a cycle. Symbolic regression was used only outside the loop for calibration and analysis; it is not part of the online discovery policy. The LLM-assisted work reported here occurs after runs are complete: it helps design follow-up experiments, interpret artifacts, and write documentation. This separation is important because every accepted law signature in the atlas can be reproduced from deterministic scripts.

4. Seven Cycle Laws

The seven laws are the primary evidence units linking raw ECA frames to the world categories, fragility scores, and observer artifacts reported in Sections 5-8. Each law is evaluated per run — one world, one initial condition, one step count — once the observer and dedup pipeline reports `analysis_status = ok`. The output is binary: accepted or rejected. Law signatures are frozensets of accepted law names.

4.1 Design rationale

The seven laws were chosen to span distinct aspects of CA behavior using measurements available from a single fixed-length run. They divide into two groups by input type:

- **Structure-observer laws** (`velocidad_constante`, `periodicidad`, `densidad_estable`, `tipo_unico`): depend on the output of the heuristic observer stack. They can only fire if the run is analyzable (`analysis_status = ok`) and the observers detect at least one structure.
- **Frame-metric laws** (`complejidad_alta`, `frontera_temporal`, `temporal_scale_stability`): depend only on summary statistics computed directly from the frame array, without reference to individual structures.

This split is intentional. Frame-metric laws are symmetry-agnostic: they fire regardless of where structures are in the lattice, how many there are, or whether the observers identify them correctly. Structure-observer laws are richer but carry the observer's heuristic assumptions. Section 8 reports two cases where structure-observer laws produce artifacts that frame-metric laws do not.

All laws produce binary output (accepted / rejected). The choice of binary rather than continuous output is also intentional: it makes signatures comparable across runs and worlds without requiring score normalization, and it forces explicit calibration of each threshold.

4.2 Formal criteria

#	Law	Inputs	Criterion	Constants
1	`velocidad_constante`	Position tracks of moving structures	At least 50% of moving tracks (`velocity > 0.05` cells/step) have linear `x(t)` with normalized residual `< 0.15`	-
2	`periodicidad`	Structure type list	At least one structure classified as `oscilador`	-
3	`densidad_estable`	Frame density time series	Coefficient of variation `CV = sigma(rho) / mu(rho) < 0.15`	-
4	`tipo_unico`	Structure type set	Exactly one structure type present	-
5	`complejidad_alta`	Frame metrics	`entropy_mean > 0.80` and `transition_rate > 0.25`	-
6	`frontera_temporal`	Frame metrics	`entropy_mean > 0.80` and `0.28 < transition_rate < 0.4352`	upper threshold calibrated 2026-05-24
7	`temporal_scale_stability`	Frame metrics + steps	`temporal_load = steps * gzip_ratio / transition_rate < 19.03`	threshold calibrated 2026-05-24

`temporal_scale_stability` rejects any run with `transition_rate = 0` (quiescent or static configurations), since temporal load is undefined (`infinity`).

4.3 Calibrated constants

Neither threshold can be derived analytically: the boundary between organized frontier dynamics and pure chaos has no closed form in ECA. Both constants were set empirically on real ECA runs and are valid within the atlas protocol (`width = 64`, `steps` roughly `24..200`).

The `frontera_temporal` upper threshold `0.4352` is the midpoint between the maximum `transition_rate` observed for `rule_110` (`0.4147`) and the minimum for `rule_30` (`0.4557`) across six canonical seeds at `steps = 24`, `width = 64`.

The `temporal_scale_stability` threshold `19.03` was fit on `datasets/fase2c_v3.csv` (120 ECA scale samples). A decision tree at `max_depth = 4` achieved accuracy `0.908`, precision `0.886`, and recall `0.954` on the `analysis_ok` label.

4.4 Caveats

`tipo_unico` is an observer-dependent exploratory signal, not a mirror-invariant physical property. Fase 6b showed that `rule_110` and `rule_124` are left-right mirrors of each other with identical dynamics, yet `tipo_unico` can fire asymmetrically depending on orientation. `tipo_unico` is retained in the atlas for its exploratory value but should not be used as evidence of physical asymmetry.

`frontera_temporal` and `temporal_scale_stability` both depend on `transition_rate`. Fase 4a and later tree analyses identify transition rate as the main discriminator separating organized frontier dynamics from pure chaos or static order. Other metrics (`density_mean`, `gzip_ratio`, `mutual_info_mean`) are useful context features but should not be treated as independent causal evidence without ablation.

These caveats do not weaken the atlas: they clarify which signals reflect physical ECA dynamics and which reflect the current observer design. Section 8 returns to both artifacts with controlled experiments.

4.5 Law signatures and the atlas

A law signature is a frozenset of accepted law names for one run. The empty frozenset is valid and indicates a run that passed the noise gate but accepted no law. Law signatures are the unit of evidence in the atlas: the world categories in Section 5 are defined by how signatures distribute across seeds and scales, and the fragility measurements in Section 6 count how often signatures change under perturbation.

`frontera_temporal` is a proper subset of `complejidad_alta` by construction (it adds the upper bound on transition rate). Any run that accepts `frontera_temporal` also accepts `complejidad_alta`; the converse is not required. This containment is visible in the law coverage matrix: every ✓ in the `frontera_temporal` column co-occurs with a ✓ in the `complejidad_alta` column.

5. World Atlas: 20 Worlds and Dynamic Categories

The atlas is derived from `outputs/world_taxonomy/law_map.md`. It contains 20 worlds: ECA rules, designed synthetic controls, and Life-like controls. Each world is summarized by law coverage, non-empty visit ratio, noise ratio, signature diversity, mean law count, dominant signature, and measured fragility where available.

The atlas is not a score table. A high `mean_laws` value, high signature diversity, and low fragility mean different things. The taxonomy therefore separates five positive dynamic families from two bookkeeping categories: `noise-bounded` for worlds stopped by the dedup gate, and `sin-evidencia-multiregimen` for controls or worlds without sufficient evidence for one of the positive families.

5.1 Category definitions

category	operational signal	representative worlds
`frontera-rich-estable`	low signature diversity, high stable law richness (`mean_laws >= 4.0`)	`rule_46`, `rule_208`, `rule_209`
`periodicidad-global`	global period-2 behavior; `periodicidad` in nearly all non-empty visits	`rule_51`
`oscilador-local`	bounded local period-2 structure on a quiescent background	`rule_108`
`multiregimen-productivo`	multiple non-empty law signatures with productive visits	`rule_18`, `rule_54`, `rule_109`, `rule_110`, `rule_124`, `rule_137`
`multiregimen-escala-dependiente`	real signature diversity but most high-scale visits become analyzable silence	`rule_90`
`noise-bounded`	pre-law failure under the deduplicated structure gate	`rule_30`, `rule_150`
`sin-evidencia-multiregimen`	no sufficient evidence of multi-regime or stable-rich behavior in the current protocol	`life_blinker`, `life_block`, `life_glider`, `synthetic_bloque`, `synthetic_glider`, `synthetic_oscilador`

world	category	mean_laws	peak_diversity	f_total	f_core
`rule_208`	`frontera-rich-estable`	6.000	0.167	0.000	0.000
`rule_209`	`frontera-rich-estable`	6.000	0.167	0.000	0.000
`rule_46`	`frontera-rich-estable`	5.833	0.333	0.031	0.031
`rule_51`	`periodicidad-global`	4.500	0.333	0.193	0.000
`rule_108`	`oscilador-local`	2.000	0.167	0.992	0.047
`rule_90`	`multiregimen-escala-dependiente`	0.500	0.600	0.172	0.000
`rule_110`	`multiregimen-productivo`	2.727	0.600	0.323	0.198
`rule_124`	`multiregimen-productivo`	2.167	0.600	0.224	0.083
`rule_109`	`multiregimen-productivo`	2.000	0.667	0.307	0.307
`rule_18`	`multiregimen-productivo`	2.308	0.800	0.349	0.135
`rule_137`	`multiregimen-productivo`	2.867	0.833	0.630	0.312
`rule_54`	`multiregimen-productivo`	1.917	0.800	0.714	0.677
`rule_30`	`noise-bounded`	1.100	0.000	0.021	0.021
`rule_150`	`noise-bounded`	0.750	0.000	0.023	0.023
`life_blinker`	`sin-evidencia-multiregimen`	3.000	0.200	1.000	1.000

world	category	mean_laws	peak_diversity	f_total	f_core
`life_block`	`sin-evidencia-multiregimen`	2.000	0.200	0.016	0.016
`life_glider`	`sin-evidencia-multiregimen`	2.357	0.333	0.032	0.032
`synthetic_bloque`	`sin-evidencia-multiregimen`	2.000	0.200	n/a	n/a
`synthetic_glider`	`sin-evidencia-multiregimen`	3.167	0.400	n/a	n/a
`synthetic_oscilador`	`sin-evidencia-multiregimen`	2.286	0.400	n/a	n/a

This classification is intentionally operational. A world can be reclassified if a wider protocol produces different evidence; the atlas records what the current deterministic protocol has measured.

The same principle applies to `frontera_temporal` candidates. Fase 20a found many additional ECA rules that are rich in `frontera_temporal` at the fixed sweep scale (`steps = 24`), but Fase 20b showed that the strongest four new candidates do not remain `frontera-rich-estable` under long-journal policy scaling. Fase 20c therefore treats `frontera-short-scale` as a candidate tier, not as an atlas-grade category. The atlas promotes worlds only when short-scale richness survives the broader validation protocol.

5.2 Law coverage

The law coverage matrix uses seven columns:

```
velocidad_constante
periodicidad
densidad_estable
tipo_unico
complejidad_alta
frontera_temporal
temporal_scale_stability
```

Each cell has one of four states: accepted in the dominant signature or in at least half of non-empty visits (`✓`), observed but below half (`·`), never observed in non-empty visits (`-`), or unknown because no non-empty visits exist (`?`).

Cell states:

- `✓`: law appears in the dominant signature or in at least 50% of non-empty visits.
- `·`: law appears in at least one non-empty visit but in less than 50%.
- `-`: non-empty visits exist and the law never appears.
- `?`: no non-empty visits.

world	vel	per	den	tipo	compl	front	tss
`life_blinker`	-	✓	✓	✓	-	-	-
`life_block`	-	-	✓	✓	-	-	-
`life_glider`	·	-	✓	✓	-	-	-
`rule_108`	-	✓	-	✓	-	-	-
`rule_109`	-	-	✓	·	✓	✓	✓
`rule_110`	✓	-	✓	-	✓	✓	·
`rule_124`	·	-	✓	-	✓	✓	✓
`rule_137`	·	-	✓	·	✓	✓	✓
`rule_150`	-	-	✓	-	✓	-	✓
`rule_18`	✓	-	-	✓	✓	-	✓
`rule_208`	✓	-	✓	✓	✓	✓	✓
`rule_209`	✓	-	✓	✓	✓	✓	✓
`rule_30`	-	-	✓	-	✓	-	✓
`rule_46`	✓	-	✓	✓	✓	✓	✓
`rule_51`	-	✓	✓	✓	✓	-	✓
`rule_54`	✓	-	·	·	✓	-	✓
`rule_90`	·	-	·	-	·	-	✓
`synthetic_bloque`	-	-	✓	✓	-	-	-
`synthetic_glider`	✓	-	✓	✓	-	-	·
`synthetic_oscilador`	-	✓	-	✓	-	-	·

The matrix reveals three broad patterns:

1. **Synthetic and Life-like controls validate observer semantics.** `life_blinker` and `synthetic_oscilador` activate `periodicidad`; block-like worlds activate `densidad_estable` and `tipo_unico`; synthetic gliders activate `velocidad_constante`.
2. **Class-4 and frontier worlds separate into distinct families.** `rule_137`, `rule_110`, `rule_124`, `rule_109`, and `rule_54` are multi-regime worlds with two or three dominant laws. By contrast, `rule_46`, `rule_208`, and `rule_209` activate six of seven laws with low diversity.
3. **`periodicidad` is IC-family sensitive.** Under random ICs it appears in designed controls, in the global complement rule (`rule_51`), and in the local oscillator (`rule_108`), but not in the complex frontier worlds. Under explicitly periodic ICs, however, Fase 21a finds production `periodicidad` in `207/256` ECA rules. The law is therefore not dead or ECA-inaccessible; it is controlled by the IC family. This is why Section 7 treats `rule_108` separately rather than folding it into ordinary stable-rich behavior.

5.3 Key atlas rows

The following rows anchor the category structure:

world	category	mean_laws	peak_diversity	dominant signature
`rule_208`	`frontera-rich-estable`	`6.000`	`0.167`	`velocidad_constante + densidad_estable + tipo_unico + complejidad_alta + frontera_temporal + temporal_scale_stability`
`rule_209`	`frontera-rich-estable`	`6.000`	`0.167`	`velocidad_constante + densidad_estable + tipo_unico + complejidad_alta + frontera_temporal + temporal_scale_stability`
`rule_46`	`frontera-rich-estable`	`5.833`	`0.333`	`velocidad_constante + densidad_estable + tipo_unico + complejidad_alta + frontera_temporal + temporal_scale_stability`
`rule_137`	`multiregimen-productivo`	`2.867`	`0.833`	`densidad_estable + complejidad_alta + frontera_temporal`
`rule_54`	`multiregimen-productivo`	`1.917`	`0.800`	`complejidad_alta + temporal_scale_stability`
`rule_51`	`periodicidad-global`	`4.500`	`0.333`	`periodicidad + densidad_estable + tipo_unico + complejidad_alta + temporal_scale_stability`
`rule_108`	`oscilador-local`	`2.000`	`0.167`	`periodicidad + tipo_unico`
`rule_90`	`multiregimen-escala-dependiente`	`0.500`	`0.600`	`temporal_scale_stability`

These rows show why the taxonomy cannot be reduced to a single richness score. `rule_208` and `rule_209` are maximally rich and stable; `rule_137` is less rich but highly diverse; `rule_108` is law-sparse but category-defining because it is the only local oscillator; `rule_90` has high diversity evidence but low non-empty yield because its high-scale visits become silent.

5.4 Scientific role of the atlas

The atlas is the bridge between cycle-level laws and world-level claims. A law signature describes one run. A world category describes how signatures behave across seeds, scales, and perturbations. This distinction is what makes later fragility measurements interpretable: `f_total = 0.630` in `rule_137` means something different from `f_total = 0.992` in `rule_108` because the atlas identifies different category-defining cores.

6. Fragility: `f_total`, `f_core`, `f_gap`

6.1 Protocol

Fragility is measured by exhaustive one-bit IC perturbation. For each measured world and canonical seed, every bit in the IC is flipped individually, and the resulting run is evaluated through the full pipeline (simulation, frame metrics, observers, dedup, law evaluation). The reference is the law signature of the unperturbed run.

Protocol parameters:

- **IC width**: `64` for most fragility measurements; `128` for the designed `rule_108` local-oscillator IC.
- **Perturbations per seed**: one per bit position (`64` or `128`, depending on IC width).
- **Seeds per world**: usually `3` canonical seeds, giving `192` perturbations for width-64 worlds. Designed-IC worlds such as `rule_108` use a canonical IC rather than random seeds.
- **Steps**: world-specific canonical steps (e.g., `24` for `rule_46`, `48` for `rule_137`, `96` for `rule_54`).

6.2 Metrics

Three primary fragility metrics are defined:

- **`f_total`**: fraction of perturbations that produce a different law signature from the reference (including noise-gated runs and silence).
- **`f_core`**: fraction that changes the category-defining core laws. Noise and silence count as core changes because the defining regime is lost.
- **`f_gap = f_total - f_core`**: secondary-law churn. Perturbations that change the signature without affecting the core-defining laws.

A fourth component is tracked separately:

- **`f_noise`**: fraction of perturbations that produce `analysis_status = ruido_no_analizable` (noise-gate crossing).

`f_noise` is a component of `f_total` and `f_core`; it is reported separately because it identifies a specific observer-boundary mechanism.

6.3 Core-law convention

Core laws are defined per category:

category	core laws
`frontera-rich-estable`	the full six-law frontier signature
`periodicidad-global`	`periodicidad`
`oscilador-local`	`periodicidad` and `tipo_unico`
`multiregimen-productivo`	the reference signature of that specific seed
`multiregimen-escala-dependiente`	`temporal_scale_stability`

For `multiregimen-productivo` worlds, the reference signature varies by seed. `f_core` is therefore computed per seed against that seed's reference and then averaged.

6.4 Fragility spectrum

Fase 22 completes the physical fragility spectrum for all measured non-synthetic worlds in the atlas. Synthetic controls are excluded from `f_total` and `f_core` because they are frame generators rather than evolved systems with a perturbable initial condition. The completed spectrum includes ECA worlds, Life fixtures, stable-rich frontier worlds, noise-bounded productive pockets, and the designed `rule_108` local-oscillator IC.

world	category	`f_total`	`f_core`	`f_gap`	`f_noise`	mechanism
`rule_208`	`frontera-rich-estable`	0.000	0.000	0.000	0.000	stable basin
`rule_209`	`frontera-rich-estable`	0.000	0.000	0.000	0.000	stable basin
`life_block`	`sin-evidencia-multiregimen`	0.016	0.016	0.000	0.000	stable Life fixture
`rule_30`	`noise-bounded`	0.021	0.021	0.000	0.000	productive pocket
`rule_150`	`noise-bounded`	0.023	0.023	0.000	0.000	productive pocket
`rule_46`	`frontera-rich-estable`	0.031	0.031	0.000	0.000	stable basin
`life_glider`	`sin-evidencia-multiregimen`	0.032	0.032	0.000	0.000	stable Life fixture
`rule_90`	`multiregimen-escala-dependiente`	0.172	0.000	0.172	0.000	secondary churn
`rule_51`	`periodicidad-global`	0.193	0.000	0.193	0.000	secondary churn
`rule_124`	`multiregimen-productivo`	0.224	0.083	0.141	0.000	productive switching
`rule_109`	`multiregimen-productivo`	0.307	0.307	0.000	0.000	productive switching
`rule_110`	`multiregimen-productivo`	0.323	0.198	0.125	0.000	productive switching
`rule_18`	`multiregimen-productivo`	0.349	0.135	0.214	0.000	productive switching
`rule_137`	`multiregimen-productivo`	0.630	0.312	0.318	0.000	productive switching
`rule_54`	`multiregimen-productivo`	0.714	0.677	0.037	0.375	noise-boundary
`rule_108`	`oscilador-local`	0.992	0.047	0.945	0.000	quiescent-background activation
`life_blinker`	`sin-evidencia-multiregimen`	1.000	1.000	0.000	0.000	periodic fixture disruption

The spectrum is category-aligned at the extremes: `frontera-rich-estable` occupies the low end, while `multiregimen-productivo` occupies the upper ECA range. The `life_blinker` control reaches `f_total = 1.000` because any single-cell perturbation breaks the exact Life oscillator fixture. `rule_108` remains the main ECA structural outlier: `f_total = 0.992` with only `f_core = 0.047`.

6.5 Fragility mechanisms

The atlas identifies several distinct mechanisms by which one-bit IC perturbations change law signatures:

Stable basin (`rule_208`, `rule_209`): `f_total = 0.000`. All perturbations preserve the reference signature. The basin for the six-law frontier signature is wide enough that no measured single-bit perturbation escapes it. `rule_46` is nearly identical (`f_total = 0.031`).

Stable Life fixture (`life_block`, `life_glider`): perturbations rarely change the reference signature (`f_total <= 0.032`) because the fixture remains structurally recognizable after most one-cell flips. This is fixture-level robustness, not evidence of a broad ECA basin.

Productive pocket (`rule_30`, `rule_150`): the worlds are noise-bounded in the long journal, but the non-empty pockets that survive the gate are stable under one-bit perturbation (`f_total ~= 0.02`). Their category is defined by frequent pre-law noise at scale, not by fragility of the productive signatures.

Productive basin switching (`rule_137`, and the non-noise-boundary `multiregimen-productivo` worlds): perturbations move the IC among productive law-signature regimes. The world never falls into silence or noise; `f_noise = 0.000` throughout. `rule_137` is the strongest clean case (`f_total = 0.630`), with more than 80% of perturbations switching regime in the two most fragile measured seeds.

Noise-boundary fragility (`rule_54`): perturbations cross the observer noise gate rather than moving between productive regimes. The mechanism requires complex ICs near the dedup threshold; single-bit ICs from bare backgrounds do not approach the gate (Section 8). `f_noise = 0.375` makes `rule_54` the only measured world where noise-gate crossings dominate fragility.

Periodic fixture disruption (`life_blinker`): any one-cell perturbation breaks the exact Life period-2 reference signature (`f_total = 1.000`). This is not basin switching or noise-boundary crossing; it is the brittleness of a minimal periodic fixture under full-grid perturbation.

Quiescent-background activation (`rule_108`): the canonical IC has only two active bits on a zero background. Nearly any background perturbation ignites new dynamics and changes secondary laws, producing `f_total = 0.992`. The core oscillator survives unless the perturbation lands near the motif (`f_core = 0.047`). The result is the largest `f_gap` in the atlas (`0.945`): nearly all fragility is secondary, not core.

6.6 The `f_core` / `f_gap` separation

The main result of the fragility analysis is the separation between core and secondary law changes:

- `rule_51` (`periodicidad-global`): `f_total = 0.193`, `f_core = 0.000`. Global periodicity survives all measured perturbations; only secondary laws (`densidad_estable`) toggle.
- `rule_108` (`oscilador-local`): `f_total = 0.992`, `f_core = 0.047`. The local oscillator survives nearly all perturbations; secondary laws are maximally sensitive.
- `rule_54` (`multiregimen-productivo`): `f_total = 0.714`, `f_core = 0.677`. The core productive signature changes frequently; `f_gap = 0.037` is near zero.
- `rule_137` (`multiregimen-productivo`): `f_total = 0.630`, `f_core = 0.312`. Both core and secondary transitions are common; `f_gap` is approximately equal to `f_core`.

These four cases span the space of possible (`f_core`, `f_gap`) combinations. Together they demonstrate that `f_total` alone is insufficient: two worlds can have similar total fragility with opposite core/secondary decompositions.

7. Case Studies

7.1 `rule_108` — Unique Local Oscillator

Discovery and formal profile

`rule_108` was identified during a targeted local-oscillator search (Fase 16) using minimal ICs on a quiescent background (`f(0,0,0) = 0`). The canonical IC is a pair of active cells separated by one gap (`#.#`, word `101` in binary). Under `rule_108`, this IC produces an exact period-2 local oscillator (Figure 4): the gap fills in each step (`#.# -> ###`) and then empties again, repeating indefinitely with zero drift.

Figure 4. `rule_108` period-2 oscillator motif.

```
Step t:   . . . # . # . . .
Step t+1: . . . # # # . . .
Step t+2: . . . # . # . . .   (repeats)
```

Space-time diagram to be rendered as bitmap figure. Active cells shown as `#`, quiescent background as `.`. Span <= 3, period `T = 2`.

The oscillator is stationary (center of mass fixed), bounded (`span <= 3`), and stable over 200 steps with zero drift on a uniform-zero background.

Formal profile (6 canonical seed labels, `width = 128`, `steps = 200`, IC = `pair_gap1`): `periodicidad` and `tipo_unico` are accepted in `6/6` runs. Mean `dedup_structure_count = 1.000`. The oscillator is deterministic given the canonical IC; the seed labels are retained only to keep the profile format consistent with other atlas worlds.

Algebraic derivation

The oscillator follows from three entries in the `rule_108` table:

Neighborhood	Output	Role
`010`	`1`	isolated active center stays active
`101`	`1`	gap fills in: `#.# -> ###`
`111`	`0`	center empties: `### -> #.#`

`rule_108` is left-right symmetric (`f(l,c,r) = f(r,c,l)`), which explains why the oscillator does not drift: the two flanking cells exert equal influence on the center.

Fragility: quiescent-background activation

`rule_108` has the largest fragility gap in the atlas:

Metric	Value
`f_total`	`0.992`
`f_core`	`0.047`
`f_gap`	`0.945`
Core positions	`61, 62, 63, 65, 66, 67`
Pattern	`clustered`

The mechanism is geometric. The canonical IC (`pair_gap1`, 2 active bits in `width = 128`) leaves more than 120 cells at zero. A one-bit perturbation at any of those background positions activates the quiescent background: because `f(0,1,0) = 1`, an isolated `1` on a zero background immediately grows, producing new detectable structures. This changes secondary laws while leaving `periodicidad` and `tipo_unico` intact as long as the oscillator core is undisturbed. A perturbation within the core neighborhood (positions `61..63`, `65..67`) displaces or destroys the oscillator, accounting for `f_core = 0.047`.

This constitutes a third fragility mechanism, distinct from productive basin switching (`rule_137`, Section 7.3) and noise-boundary fragility (`rule_54`, Section 7.2): quiescent-background activation. The core behavior is robust, but the minimal IC makes secondary laws highly sensitive to background activation.

Uniqueness

Fase 18 ran an exhaustive search over all 128 ECA rules with quiescent backgrounds (`f(0,0,0) = 0`), testing 502 non-zero IC words per rule (all non-empty binary words of length 1..8), with `width = 128`, `steps = 200`, burn-in of 80 steps, and requiring zero drift (stationary oscillators only). Only `rule_108` produced stationary local period-2 oscillators; no other rule produced a stationary local oscillator for periods `2..16` and span `<= 32`.

A companion sweep with the stationarity requirement relaxed found a distinct family of moving local oscillators -- see Section 7.5.

The longer-IC extension strengthens the same conclusion. Extending the stationary sweep to all 7,676 non-zero IC words of length 9..12 (`982,528` rule/IC runs) produced 3,802 detections, all in `rule_108`. No new stationary oscillator rule appears beyond the length-1..8 baseline; the minimum witness is still the embedded `101` motif (`000000101` at length 9).

The family is internal to `rule_108`: 132 out of 179 candidate IC words are accepted by the production observer as `periodicidad`, with oscillator spans 3, 5, 6, 7, and 8. All confirmed oscillators have period exactly 2; no longer period was found.

7.2 `rule_54` — Noise Gate Anatomy

`rule_54` is the clearest example of noise-boundary fragility: perturbations do not merely move the run to another productive signature, but can push the observer output across the deduplicated structure gate. The production gate is:

```
dedup_structure_count > 40 -> ruido_no_analizable
```

Fase 13: anatomy of the gate

Fase 13 measured three productive `rule_54` ICs at `steps = 96` and perturbed each by all 64 one-bit flips. The reference deduplicated counts were close to the gate:

seed	reference dedup	noisy flips / 64
20260638	32	14
20260640	33	18
20260642	39	40

Across the three seeds, `72/192` flips crossed into `ruido_no_analizable` (`f_noise = 0.375`). Every noisy flip crossed for the same reason: `dedup_structure_count > 40`. No alternative noise mechanism was observed.

The sensitive positions formed a clustered, multi-hot pattern rather than a single contiguous block. Bins near the periodic boundary (`0..7` and `56..63`) were repeatedly implicated, and bit 5 was the only bit whose flip crossed the gate in all three measured seeds.

Fase 19: controlled single-bit negative case

Fase 19 tested whether bit 5 was a special absolute coordinate of `rule_54`. It replaced the complex ICs with controlled single-bit ICs: for each `k = 0..63`, the initial state contained only one active bit at position `k`.

The result separates CA physics from the observer pipeline:

- The ECA frames are translation-invariant after shift normalization.
- The observer/dedup counts are not translation-equivariant for this wide-spreading pattern: `dedup_structure_count` ranges from `15` to `24` across positions.
- The law signature is identical for all 64 positions: `temporal_scale_stability`.
- Every single-bit IC remains far below the gate (`dedup <= 24 < 40`).

Therefore, bit 5 is not a privileged coordinate of `rule_54`. The Fase 13 signal arises from the interaction between complex IC geometry and the observer/gate pipeline. Complex ICs close to the threshold can be pushed across it by local flips; a single active cell cannot.

Mechanism

`rule_54` has high total and core fragility (`f_total = 0.714`, `f_core = 0.677`), but its mechanism differs from `rule_137`. In `rule_137`, perturbations tend to move between productive regimes. In `rule_54`, a large fraction of perturbations cross an analysis boundary: the run becomes too fragmented for the current observer/dedup gate.

This makes `rule_54` a methodological case study as much as a dynamical one. It shows that the atlas can identify worlds whose measured fragility is dominated by proximity to an observer threshold. It also motivates the caveat that absolute structure counts should not be treated as symmetry-invariant physical observables without equivariance checks.

7.3 `rule_137` — Productive Basin Switching

`rule_137` is the primary example of productive basin switching: one-bit IC perturbations change the law signature without ever crossing the noise gate or reaching silence. All fragility is productive (`f_noise = 0.000`, `f_silence = 0.000`), making it the cleanest case in the atlas for inter-basin transitions.

Fragility profile

Three canonical seeds at `steps = 48`, `width = 64`:

seed	reference signature	f_total
20260633	complejidad_alta + frontera_temporal	0.812
20260635	complejidad_alta + densidad_estable + frontera_temporal	0.219
20260673	complejidad_alta + densidad_estable + frontera_temporal + temporal_scale_stability + tipo_unico + velocidad_constante	0.859

Aggregate: `f_total = 0.630`, `f_core = 0.312`, `f_gap = 0.318`.

The per-seed range (`0.219..0.859`) is the widest in the atlas. Even the least fragile measured seed has `f_total > 0.2`. The most fragile seeds (20260633 and 20260673) flip on more than 80% of one-bit perturbations.

Mechanism

`f_noise = 0.000`: no perturbed IC crosses the deduplicated structure gate. The world remains analyzable throughout. The fragility is a property of productive basin geography, not proximity to an observer threshold.

The pattern is `dispersed`: sensitive positions are distributed across the IC width rather than concentrated near a motif. This is consistent with a world that has many narrow productive basins whose boundaries intersect throughout the IC space.

`peak_diversity = 0.833` — the highest in the atlas. The canonical seeds themselves already visit multiple distinct productive regimes. The fragility measurement extends this: not just that the world can reach different signatures under different seeds, but that a single-bit perturbation to any one canonical IC is enough to move between regimes.

f_core and f_gap interpretation

`f_core = 0.312` reflects genuine regime switching: flips that remove or change laws defining the canonical signature. `f_gap = 0.318` reflects secondary-law churn: the core productive signature survives, but laws on the signature boundary (such as `densidad_estable` or `tipo_unico`) toggle.

The two components are roughly equal (0.312 vs 0.318), meaning `rule_137` sits in a region where both core-regime transitions and secondary-law transitions are common. This is structurally different from `rule_108` (`f_gap = 0.945`, where the core oscillator is robust and secondary laws dominate) and from `rule_54` (`f_gap = 0.037`, where the core productive signature changes but almost no fragility is secondary).

Contrast with `rule_54` and `rule_108`

`rule_54` and `rule_137` both have high `f_total` (0.714 vs 0.630), but the mechanisms are opposite: `rule_54` fragility is dominated by noise-gate crossings (`f_noise = 0.375`), while `rule_137` fragility is entirely productive. A perturbed `rule_137` IC stays analyzable and law-rich; it is simply in a different productive regime.

`rule_108` contrasts from the opposite direction: its `f_core = 0.047` shows that the defining behavior (the local oscillator) is nearly indestructible, while `rule_137`'s `f_core = 0.312` shows that its defining signatures change under nearly a third of all one-bit flips.

7.4 `rule_46`, `rule_208`, `rule_209` — Stable-Rich Frontier

These three worlds define the `frontera-rich-estable` category: low signature diversity, near-maximal law richness, and very low fragility. They are the counterexample that revised the early atlas interpretation of `frontera_temporal`.

The first 15-world atlas made `frontera_temporal` look rare: it appeared only as a minority law in class-4 multi-regime worlds such as `rule_137`, `rule_110`, and `rule_54`. Fase 11 showed that this was a sampling artifact. A sweep over all 256 ECA rules (`seeds = 20260523..20260525`, `W = 64`, `T = 24`) found 38 rules where `frontera_temporal` activates in at least two of three seeds, and 17 rules where it activates in all three.

The top rules by law richness were `rule_46`, `rule_208`, and `rule_209`. Formal six-seed profiles (`20260523..20260528`, `W = 64`, `T = 24`) placed all three in a new category:

world	mean laws	peak diversity	category
`rule_46`	5.833	0.333	`frontera-rich-estable`
`rule_208`	6.000	0.167	`frontera-rich-estable`
`rule_209`	6.000	0.167	`frontera-rich-estable`

The dominant signature is the same six-law set:

```
velocidad_constante
densidad_estable
tipo_unico
complejidad_alta
frontera_temporal
temporal_scale_stability
```

Only `periodicidad` is absent. This makes the family nearly maximal under the current seven-law system without relying on multi-regime exploration.

Stable richness rather than multi-regime diversity

The category is defined by the conjunction of high richness and low diversity. `rule_137` is rich because it moves among several productive law signatures. The frontier-rich worlds are rich because the same high-law signature appears reliably across seeds.

This distinction matters operationally. A policy that only looks for signature diversity would miss these worlds, even though they produce more accepted laws per visit than any multi-regime world in the atlas. The Fase 11 taxonomy update therefore adds:

```
frontera-rich-estable := mean_laws >= 4.0 and peak_diversity <= 0.5
```

evaluated after noise-bounded and multi-regime cases.

Complement symmetry and independent convergence

`rule_46` and `rule_209` are a complement pair (`46 = 255 - 209`): exchanging zeros and ones maps one into the other. Their shared profile is therefore one physical phenomenon seen through global bit inversion.

`rule_208` is more surprising. Its complement is `rule_47`, not `rule_46` or `rule_209`, yet it reaches the same maximum-richness profile. This suggests that the `frontera-rich-estable` regime is not a single isolated symmetry orbit; at least two distinct ECA regions converge to the same six-law frontier.

Fragility

Fase 12 measured one-bit fragility for the three worlds using the same protocol as `rule_137` and `rule_54`:

world	f_total	f_core	f_noise
`rule_46`	`0.031`	`0.031`	`0.000`
`rule_208`	`0.000`	`0.000`	`0.000`
`rule_209`	`0.000`	`0.000`	`0.000`

The result is the opposite end of the fragility spectrum from `rule_137`. Where `rule_137` has many narrow productive basins (`f_total = 0.630`), `rule_208` and `rule_209` have measured basins so wide that no single-bit flip changes the law signature. `rule_46` is only slightly fragile: two of 192 single-bit perturbations change signature across the three measured seeds.

This confirms that high law richness does not imply high fragility. Richness can arise either from many neighboring productive regimes (`rule_137`) or from a single broad, stable regime (`rule_46/208/209`).

Scientific revision

The correct conclusion is not that `frontera_temporal` is intrinsically rare. It is rare in the original discovery atlas because the original world sequence under-sampled stable high-richness boundary worlds. In the full ECA sweep, `frontera_temporal` is a robust marker of the `frontera-rich-estable` family.

7.5 Moving Oscillator Family -- Minimal Period-2 Gliders

A companion sweep to Fase 18 searched all 128 quiescent ECA rules (`f(0,0,0) = 0`) for oscillators that translate at constant velocity. The detection protocol matched Fase 18 except: the zero-drift requirement was replaced by a requirement of constant non-zero drift confirmed over three consecutive periods; `width` was extended to `256` and `steps` to `300` to give moving patterns room to travel. The 502 non-zero IC words of length 1..8 were tested per rule (64,256 total runs, 312 s).

Eight rules produce moving oscillators: `rule_6`, `rule_20`, `rule_38`, `rule_52`, `rule_134`, `rule_148`, `rule_166`, `rule_180`. All share the same minimal glider pattern:

Step	Active shape (normalized offsets)
`t`	`[0]` -- single active cell
`t+1`	`[0, 1]` -- two adjacent active cells
`t+2`	`[0]` displaced by `+/-2`

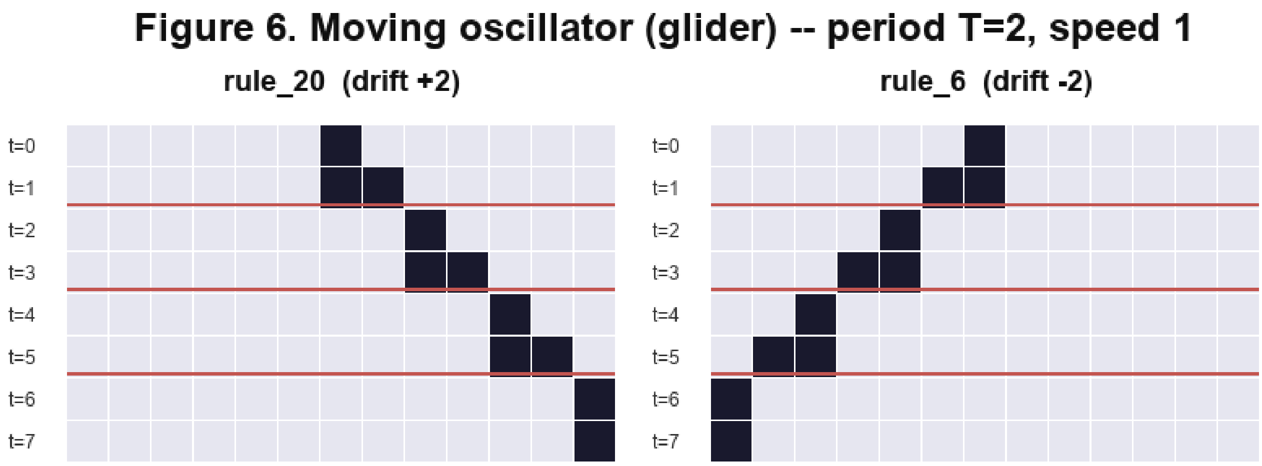
The oscillator alternates between one and two active cells while traveling at speed 1 cell per step -- the maximum velocity for a radius-1 ECA. Mean active span per period is 0.5 (alternating span 0 and span 1). All eight rules are confirmed `edge_touch = False` within `width = 256`.

Figure 6. Moving oscillator (glider) -- period T=2, speed 1.

```
rule_20 (drift +2)          rule_6 (drift -2)

t=0:  . . . 1 . . . . .    . . . . . 1 . . .
t=1:  . . . 1 1 . . . .    . . . . 1 1 . . .
t=2:  . . . . . 1 . . .    . . . 1 . . . . .   <- period boundary
t=3:  . . . . . 1 1 . .    . . 1 1 . . . . .
t=4:  . . . . . . . 1 .    . 1 . . . . . . .   <- period boundary
```

Active cells shown as `1`, quiescent background as `.`. Each period advances the pattern two positions in the travel direction.



Left: rule_20 (right-moving, drift +2 per period). Right: rule_6 (left-moving, drift -2 per period). Active cells dark; quiescent background light. Rows are time steps t=0..7. The oscillator alternates [0] <-> [0,1] while translating one cell per step.

Structure: four mirror pairs

The eight rules form four left-right symmetric pairs:

Left-moving	Right-moving	`b5`	`b7`
`rule_6`	`rule_20`	0	0
`rule_38`	`rule_52`	1	0
`rule_134`	`rule_148`	0	1
`rule_166`	`rule_180`	1	1

Bits `b5` (neighborhood `101`) and `b7` (neighborhood `111`) vary across the family but do not participate in the glider cycle: neither neighborhood occurs during quiescent travel with a two-cell active pattern. The eight rules are therefore a single physical family parametrized by two inactive bit choices and the left-right direction.

Contrast with `rule_108`

Property	`rule_108`	Moving family
Drift per period	0	+/-2
Period `T`	2	2
Mean active span	~2	0.5
Speed	0 (stationary)	1 (maximum)
Rules in scope	1	8 (4 mirror pairs)

The two families do not overlap. `rule_108` does not appear among the eight moving rules, and none of the eight moving rules produce stationary local oscillators under Fase 18. Together, the two sweeps partition the quiescent local oscillator landscape into a unique stationary oscillator (`rule_108`) and a unique minimal glider family (8 rules, 4 mirror pairs).

Uniqueness within the protocol

No IC word of length 1..8 produced a moving oscillator with `T > 2` or `|drift| != 2` in any quiescent rule. The minimal glider `[0] <-> [0, 1]` at period 2 and maximum speed is the only moving local oscillator family under this protocol.

The longer-IC extension reaches the same result. Testing all 7,676 non-zero IC words of length 9..12 over all 128 quiescent rules (`982,528` rule/IC runs) produced 2,059 moving detections, all within the same eight-rule family. No new moving-oscillator rule appears. The minimal witnesses are still the old glider seeds embedded in longer words (`000000001` for right movers, `000000010` for left movers). The sweep also filtered 9,822 period-1 moving particle aliases across 32 rules; these are moving particles, not internal period-2 oscillators. Extensions to IC words longer than 12 remain open (Section 10.2). The non-zero background extension is reported in Section 7.6.

7.6 Periodic-background oscillator sweep

The two sweeps above restrict the background to quiescent zero cells. A third sweep tests whether replacing the background with a non-zero periodic tiling changes the oscillator landscape.

Protocol. All 256 ECA rules are tested against 15 unique non-zero periodic backgrounds with template lengths 1, 2, and 4. Each rule/background pair is tested against 502 non-zero IC words of length 1..8 centered in a width-256 grid for 300 steps with 80 burn-in. The detector identifies exact recurrence of the localized difference between the perturbed run and the unperturbed background orbit; global background periodicity alone is not counted as a local oscillator. Total: 1,927,680 rule/background/IC runs, 122,253 candidate detections.

Result. Thirty rules produce stationary local oscillators under at least one non-zero periodic background; 36 rules produce moving oscillators. Of the 30 stationary rules, 29 are new relative to the zero-background baseline. Of the 36 moving rules, 28 are new.

The following phenomena appear under non-zero backgrounds but not under the quiescent zero background:

- **Period-4 stationary oscillators.** `rule_54` and `rule_147` produce stationary period-4 oscillators under `0001` background.
- **Period-4 moving oscillator.** `rule_180` produces a T=4 glider with drift +4 under `0001` background (shapes `[0] -> [0,1] -> [0,2,3] -> [0]`), distinct from its T=2 speed-1 glider under quiescent background.
- **Speed-0.5 gliders.** Multiple rules (including `rule_3`, `rule_17`, `rule_27`, `rule_35`, `rule_39`) produce T=2 gliders with drift +/-1 under non-zero backgrounds, corresponding to 0.5 cells per step. Under quiescent zero background the only observed glider speed was 1 cell/step.
- **`rule_108` under all-one background.** `rule_108` appears in the stationary list with background `1` and the same motif `### / #.#` as the quiescent result, confirming that the rule_108 oscillator is intrinsic to the rule table.

Separation of regimes. The zero-background uniqueness claims (Sections 7.1 and 7.5) are not contradicted: they describe the quiescent regime. Under zero background, the quiescent local oscillator space is sparse (1 stationary rule, 8 moving rules); under non-zero periodic backgrounds, the landscape expands substantially (30 stationary, 36 moving) and includes period and speed classes absent from the quiescent regime. The two regimes should not be merged.

7.7 Period-8 background oscillator sweep (Fase 24)

A fourth oscillator sweep extended the periodic-background protocol to length-8 primitive binary backgrounds, testing three questions: do longer background periods introduce new oscillator rules, new period classes T>4, or glider speeds outside {0, 0.5, 1}?

Protocol. All 256 ECA rules are tested against 30 primitive binary necklaces of length 8. A length-8 binary string is primitive if its minimal period is exactly 8; the necklace representative is the lexicographically smallest member of its rotation class. The count of 30 follows from Mobius inversion: `(2^8 - 2^4) / 8 = 30`. Each rule/background pair is tested against 502 non-zero IC words of length 1..8 in a width-256 grid, 300 steps, 80 burn-in. The differential detector and period search window (2..16, span <= 32) are unchanged from Section 7.6. Total: 3,855,360 runs.

Result. The sweep produces 323,872 candidate detections after filtering 95,121 period-1 aliases. New stationary rules beyond the length-1/2/4 baseline are `rule_62`, `rule_118`, `rule_131`, and `rule_145` (4 rules, all T=3). New moving rules are `rule_7`, `rule_9`, `rule_21`, `rule_25`, `rule_31`, `rule_45`, `rule_61`, `rule_65`, `rule_67`, `rule_75`, `rule_87`, `rule_88`, `rule_89`, `rule_101`, `rule_103`, `rule_111`, `rule_125`, `rule_173`, and `rule_229` (19 rules).

New period classes. Periods T=6, 8, 10, 12, and 15 appear for the first time; under backgrounds of length 1, 2, and 4 the maximum observed period was T=4. T=15 is a non-trivial period not divisible by the background length (8). Fase 24 reports it as an emergent period; Sections 7.8 and 7.9 subsequently identify its rule family and establish its five-state mechanism under `F^3`.

New glider speed. Speed 2/3 cell/step (drift +/-2, T=3) is observed for the first time; the prior speed set was {0, 0.5, 1}. Representative cases are `rule_9` (drift -2, T=3, background `00001001`) and `rule_65` (drift +2, T=3, background `00000001`). Two further rules with the same speed signature, `rule_111` and `rule_125`, appear in the candidate table. Direct rule-table reflection, `g(l,c,r) = f(r,c,l)`, confirms two exact left-right mirror pairs: `rule_9 <-> rule_65` and `rule_111 <-> rule_125`. The paired rules carry opposite drift signs with the same T=3 speed magnitude, giving the speed-2/3 family the same left/right mirror structure as the speed-1 family in Section 7.5.

Phase dependence. A rotation sub-test applied all 8 rotations of the canonical background to 10 sampled rules while holding the IC fixed. None of the 10 samples is active in all eight rotations after circular-geometry correction. `rule_62` and `rule_118` activate in 7 of 8 background rotations; moving cases range from 6/8 (`rule_9`, `rule_65`) to 2/8 (`rule_111`, `rule_45`). The IC therefore requires a particular alignment with the background to nucleate the oscillator.

Co-translation test (Fase 25). A strict test co-translates both background and IC through `k=0..7` for the same 10 cases. The background and XOR perturbation orbits are exact translations in 80/80 runs. The original `linear_shape` preprocessing recovers only 58/80 signatures: 22 moving runs cross positions 255/0, causing a false linear span and rejection. Circular shape canonicalization (largest-gap cut plus continuous position unwrapping) recovers 80/80 signatures and all 10/10 cases. Thus the physics and recurrence detector are co-translation equivariant once cyclic geometry is represented correctly. The remaining fixed-IC phase dependence is physical, although the linear observer overstated its severity for moving rules.

Summary. All three Fase-24 questions are answered affirmatively: period-8 backgrounds introduce 4 new stationary rules and 19 new moving rules, expand the period set to include T=6, 8, 10, 12, and 15, and introduce speed 2/3 cell/step as a new rational class. The zero-background uniqueness claims of Sections 7.1 and 7.5 are unaffected.

7.8 Anatomy of the T=15 family (Fase 26)

The longest period observed in Fase 24 was analyzed separately to determine whether it represented a single accidental witness or a coherent family. All 221 `T=15` detections are stationary and occur in only two rules: `rule_73` (123 detections) and `rule_109` (98). Both rules are left-right symmetric, and black/white conjugation maps each rule exactly to the other. The detections cover 14 primitive length-8 backgrounds, 20 rule/background pairs, and 25 temporal motifs up to cycle phase. The minimum witness is `rule_109` on background `00011001` with the two-cell IC word `01`.

Temporal locking. Every participating unperturbed background enters a temporal orbit of period `T_bg=3` (with transient length 0..2). The localized perturbation has fundamental period `T_local=15`, giving the same locking ratio `T_local/T_bg=5` in all 20 rule/background pairs. Thus `T=15` is not inherited directly from the spatial background length 8. Long-horizon reruns of one minimal witness per pair preserve exact recurrence through step 900 in 20/20 cases; the detector scans upward from period 1 and therefore excludes smaller fundamental periods.

Basin width. Persistence in time does not imply robustness to initialization. Holding the IC fixed while rotating the background preserves `T=15` in only 23/160 runs. One-bit mutations of the 20 minimal IC witnesses preserve it in 4/134 runs. Most failed perturbations settle into shorter localized periods `T=3` or `T=6`; a small number produce `T=12` or no localized period in the search window. The family is therefore temporally exact but basin-narrow.

These results establish a persistent background-locked family rather than a single numerical coincidence. Fase 26 measures but does not explain the five-to-one locking ratio; Fase 27 addresses its finite-state mechanism next.

7.9 Five-state locking mechanism (Fase 27)

Fase 26 measured `T_local/T_bg=5` across all 20 minimal `T=15` representatives but left the origin of that ratio open. Fase 27 closes the computational half of the question by tracking the localized XOR defect `D(t) = X(t) XOR B(t)` once per background period.

Protocol. After burn-in, the defect is sampled at `t = 81 + k*T_bg` for `k=0..20`, covering four complete candidate five-cycles for each of the 20 minimal `(rule, background, IC)` witnesses from Fase 26. Acceptance requires eight simultaneous checks: the background returns to the exact same phase at every sample; the first five defect states are mutually distinct; the fifth transition closes the cycle; no shorter cycle under `F^3` explains the sequence; four consecutive cycles repeat in both canonical and raw-position encodings; transitions remain deterministic across cycles; and the oscillator is stationary over every local period.

Results. All eight checks are satisfied for 20/20 representatives. Every `T=15` oscillator in the family is stationary (`drift=0`). The defect cycles through exactly five distinct states under `F^3`, the three-step evolution operator, and five applications of `F^3` are necessary and sufficient to restore the complete background-plus-defect configuration. The same result holds over four consecutive cycles in both relative-shape and absolute-position encodings.

Interpretation. The locking ratio `T_local/T_bg=5` is the cycle length of the defect under `F^3`, not a resonance with the spatial background length 8. The result is a computational state-cycle derivation: it establishes the finite-state mechanism without reducing the five nodes to a closed-form symbolic identity over the rule-table algebra of `rule_73/rule_109`. That symbolic derivation remains open.

7.10 Induced defect rule and exact conjugation (Fase 28)

Fase 28 examines the local algebra inside the five-state cycle. A localized XOR defect does not evolve under the original ECA rule `f` in isolation. If `b` is a three-bit background neighborhood and `d` its XOR-defect neighborhood, the exact induced update is

`delta_f(b,d) = f(b XOR d) XOR f(b)`.

The analysis profiles this induced rule over all 300 microsteps contained in the 100 `F^3` edges from the 20 minimal representatives.

Analytical conjugation. Let `C` denote bitwise complementation. Since `rule_109` is the black/white conjugate of `rule_73`, their global maps satisfy `F_109(C(X)) = C(F_73(X))`. If both the full state and background are complemented, induction gives `X_109(t)=C(X_73(t))` and `B_109(t)=C(B_73(t))` for every `t`. Therefore:

`D_109(t) = C(X_73(t)) XOR C(B_73(t)) = D_73(t)`.

At local level, the equivalent identity is `delta_109(C(b),d) = delta_73(b,d)`. Thus the defect orbit is invariant, not complemented, under the simultaneous black/white conjugation. This statement is analytical; exhaustive checks over all 64 local `(b,d)` combinations and 10/10 complemented orbit pairs serve as implementation validation.

Sparse-support hypothesis rejected. Every one of the 100 `F^3` macro-transitions uses all eight ordinary truth-table entries in its causal defect cone. No single induced `(b,d)` key appears in every macro-transition. The five-cycle therefore cannot be explained by one fixed sparse subset of local entries.

The negative result still exposes phase structure. The sizes of the induced-key intersections for transitions `S0->S1` through `S4->S0` are `[9,6,7,5,4]` for `rule_73` and `[11,11,10,9,8]` for `rule_109`. These signatures are rule-specific rather than universal. A closed-form derivation must therefore encode spatial phase or a higher-order block state, rather than truth-table entry presence alone.

7.11 Block-locality limits and background-indexed shape families (Fase 29–30)

Section 7.10 concludes that a closed-form derivation of the five-cycle must encode spatial phase rather than truth-table entry presence alone. Fase 29 and Fase 30 test two complementary refinements of that conclusion.

Defect-shape locality (Fase 29). For each of the 20 minimal representatives and each of the five cycle phases, the canonical XOR-defect shape is computed at sample times `t = 81, 84, 87, 90, 93`. If the T=15 cycle were a pure defect-only dynamic, every background would produce the same canonical shape in each phase. Instead, `rule_73` produces 8–9 distinct shapes per phase across its 10 backgrounds, and `rule_109` produces 8. Extending the comparison window outward to fixed local blocks (active defect span plus up to three padding cells on each side) does not help: no nontrivial block signature is shared across all backgrounds in any phase. The `W=0` result in the block-signature scan shows that the active defect span itself already carries background-dependent context — no additional padding is needed before the discrimination occurs. Only the trivial token `d000->0` (background cells evolving to background under all three microsteps) is universal.

Shape families (Fase 30). Although the defect shapes are background-dependent, they are not unstructured. Treating two five-state cycles as equivalent when one is a cyclic phase rotation of the other, the 20 representatives decompose into 7 distinct families for `rule_73` and 8 for `rule_109`, yielding 13 global families. The largest family has 3 members. Two families are shared across the conjugate rules: one of size 3 (two `rule_73` backgrounds and one `rule_109` background) and one of size 2 (background word `00110101` producing phase-rotationally equivalent defect cycles under both rules). This second coincidence is not implied by the analytical conjugation of Fase 28 — the bitwise complement of `00110101` is `11001010`, a distinct word — so it is an independent structural coincidence. Scalar background descriptors (`active_count`, `transition_count`, `active_transition_pair`) do not determine the family. The canonical temporal orbit of the background under the same rule is exact in this representative set, but this is largely a restatement of full background identity rather than a compact symbolic law.

Interpretation. Fase 29 and Fase 30 together bracket the derivation problem: a correct symbolic account must use the full temporal background orbit (or an equivalent compact encoding), and maps that orbit to one of a finite set of defect-cycle shapes plus a phase offset. A derivation predicting a single universal five-state cycle is falsified by Fase 29; a derivation predicting arbitrary unclustered shape variation is refuted by Fase 30.

7.12 Compact state variable for the T=15 family (Fase 31–32)

Fase 30 reduced the T=15 family to 13 shape families but required the full temporal background orbit as the discriminating descriptor. Fase 31 and Fase 32 search for a shorter description.

Descriptor search (Fase 31). A compact global background descriptor does not exist among the candidates tested (length-2..4 circular subpattern counts, parity, run lengths, and orbit prefixes up to 24 bits). Globally, only `orbit_prefix_24` determines the family, which is substantially a restatement of full background identity. Conditioned on the ECA rule, the shortest non-orbit candidate is the circular multiset of length-4 background subwords (`subpatterns_len4`): it separates all 10 backgrounds per rule into distinct buckets with zero ambiguity. A global decision tree over all features achieves only 0.700 training accuracy at depth 4, confirming that no shallow feature combination separates all 13 families simultaneously.

Rotation generalization (Fase 32). The circular subpattern multiset is invariant under background rotation. Fase 32 tests whether the predicted family is preserved across all seven non-trivial rotations of each of the 20 representative backgrounds, under two modes: `fixed_ic` (background rotated, IC position fixed) and `cotranslated_ic` (background rotated, IC shifted by the same amount to preserve local alignment).

Under `fixed_ic`, only 3 of 140 rotations produce T=15 at all, and 1 matches the predicted family. Under `cotranslated_ic`, all 140 of 140 rotations produce T=15 and all 140 match the predicted family. The co-translation result is not evaluated on new backgrounds but on rotational variants of the known 20 representatives; it validates that the descriptor is rotation-equivariant within the confirmed family set, not that it predicts previously unseen data.

Compact state variable. The complete minimal description consistent with all observations is the triple `(rule, subpatterns_len4, IC/background alignment)`. The rule identity determines which of the two conjugate families applies; the subpattern multiset identifies the shape-family class within that rule; and the IC/background alignment selects the phase within the five-state cycle. A derivation operating on the background word alone without IC alignment is falsified by the `fixed_ic` mode (3/140 detections). Any derivation that correctly maps this triple to a defect cycle shape and phase offset is a complete symbolic account of the T=15 family.

7.13 External length-9/10 validation of the T=15 mechanism (Fase 33-34)

Fase 33 first audits whether the compact descriptor from Section 7.12 can be falsified inside the length-8 background universe. Across all binary circular length-8 backgrounds there are only two collisions of the `subpatterns_len4` descriptor: `00110111/00111011` and `00010011/00011001`. Both collisions are already inside the confirmed T=15 set and both preserve the same defect-cycle family under the corresponding rule. There is no unseen length-8 background, outside rotations of the 20 known representatives, that shares a T=15 `subpatterns_len4` descriptor under the same rule. Thus the natural external test of the descriptor is impossible inside length 8.

Fase 34 therefore moves outside the length-8 universe, but only after a preflight checks whether the prerequisite for the five-to-one locking mechanism still exists. Primitive length-9 and length-10 backgrounds do contain temporal period-three cases under both `rule_73` and `rule_109`: 11 backgrounds per rule at length 9 and 22 per rule at length 10. A targeted validation then tests only these 66 backgrounds, the two rules, and the same 502 non-zero IC words of length 1..8, for 33,132 rule/background/IC runs.

The targeted sweep finds 90 T=15 detections across 8 external backgrounds: one length-9 background under `rule_73`, five length-10 backgrounds under `rule_73`, and two length-10 backgrounds under `rule_109`. No length-9 background under `rule_109` produces a T=15 witness in this test. These backgrounds are not rotations of the length-8 representatives. The result therefore shows that the T=15 mechanism is not an artifact of primitive length-8 backgrounds: it generalizes when `T_bg=3` is preserved. The compact descriptor from Section 7.12 remains a length-8 family identifier; extending that descriptor to variable background length remains a separate symbolic problem.

7.14 Transition tables and effective orbit embeddings (Fase 35-38)

Sections 7.10--7.13 reduce the `T=15` family to finite defect-cycle shapes, compact length-8 descriptors, and external `T_bg=3` witnesses. Fases 35--38 then move from visual shape families to the explicit macro-operator acting on the localized defect.

Transition tables (Fase 35). For each of the 20 minimal representatives, the five transitions `D_i -> D_{i+1}` under `F^3` are expanded into explicit local transition tables over the induced defect rule `delta_f(b,d)=f(b XOR d) XOR f(b)`. The table signature is never shared across different visual families, so it is a sufficient discriminator. It is not identical to the visual family partition, however: the verdict is `TABLE_REFINES_FAMILY`. Some multi-member families share the same table exactly or up to cyclic phase rotation, while others split by rule or by mechanism. This shows that the fundamental object is the induced macro-transition table, with the visual family as a coarser quotient.

Effective orbit identity (Fase 36). The clearest table identity appears in family `F00` under `rule_109`: the backgrounds `00001001`, `00010011`, and `00011001` have the same exact five-phase transition table. Fase 36 explains this algebraically at the orbit level. Although the three initial backgrounds have different preperiods, by the sampling time they have converged to the same canonical period-3 background orbit `{00001001, 00101101, 00111111}`. The shared table is therefore not accidental: after burn-in, the defect sees the same sequence of background states.

Canonical orbit insufficiency (Fase 37). Generalizing the F00 explanation to all representatives reveals a sharper constraint. Across the confirmed `T=15` set there is only one canonical period-3 background orbit for `rule_73` and one for `rule_109`; this orbit identity is too coarse to determine the 13 shape families. The missing variable is not which orbit is reached, but how the localized defect is embedded into that orbit.

Embedding descriptor (Fase 38). Fase 38 tests descriptors based on the sampled orbit step, spatial rotation offset, IC start, IC length, defect anchor, and the first sampled defect state. The first sufficient descriptor is `(rule, sample_orbit_step, sample_rotation_offset, defect_state0)`. This closes the post-burn-in description: once the defect has crystallized into its first stable-cycle state, the family is determined. The descriptor is not yet a closed-form prediction from the initial condition, because `defect_state0` is measured after burn-in.

7.15 Pre-burn-in entry phase limits (Fase 39)

Fase 39 tests the remaining left side of the causal chain:

`(background, IC) -> burn-in -> defect_state0 -> family`.

For each of the 20 minimal `T=15` representatives, the localized defect `D(t)=X(t) XOR B(t)` is traced every three ECA steps from `t=0` to the sampling time `t=81`. The first time at which `D(t)` belongs to one of the five stable cycle states is recorded as the entry time, together with the corresponding entry phase.

The entry is always fast: observed entry times are `0, 3, 6, 9, 12`, with 15/20 representatives entering at `t=3` and all 20 entering by `t=12`. However, fast entry is not the same as compact predictability. The descriptor tests include `rule+IC`, `rule+IC length`, post-hoc entry-time descriptors, and local background/IC windows of radius 1..5 around the initial perturbation. Exact predictors exist, but they are not compact: `rule+IC` uses 18 buckets with 17 singleton buckets, and the local windows produce 20 singleton buckets. Post-hoc descriptors involving entry time are exact for entry phase but do not predict it from the initial state.

The verdict is `NONCOMPACT_PREBURNIN_DESCRIPTOR_FOUND`. The stable-cycle entry point is reached after only a few applications of `F^3`, but the tested pre-burn-in descriptors either fail or identify individual cases. Thus the current derivation is complete after burn-in and sharply delimited before burn-in: predicting `defect_state0` from the raw background/IC pair remains the irreducible part of the mechanism under the tested descriptor class.

7.16 Early causal-cone compression (Fase 40)

Fase 40 converts the negative result of Section 7.15 into a constructive causal compression test. Instead of seeking a closed-form pre-burn-in descriptor, it asks whether the relevant state can be recovered by simulating only the local causal cone of the initial perturbation.

For each of the 20 minimal `T=15` representatives, the full reference remains the 256-cell simulation through `t=81`. The local predictor tests windows at `t = 3, 6, 9, 12`. Two window definitions are used: `span`, the full IC span plus a causal margin of `t` cells on each side, and `center`, the strict `2t+1` window around the IC center. Boundary cells just outside the local window are supplied by the unperturbed periodic background orbit, which is valid because a radius-1 perturbation cannot reach beyond the cone within `t` steps.

The strict center cone succeeds at `t=12`. Its 25-cell, 12-step simulation matches the full-system defect state at `t=12` in 20/20 representatives. When the detected stable-cycle phase is projected forward to `t=81`, it recovers `defect_state0` in 20/20 representatives. The compression ratio relative to the full 256-by-81 simulation is 69.1x. Earlier windows are incomplete: `center` at `t=9` reaches 17/20 stable states, while `span` at `t=9` reaches 19/20; both miss at least one representative.

The verdict is `EARLY_CONE_PREDICTOR_FOUND`. Fase 39 showed that no compact closed-form descriptor was found under the tested pre-burn-in descriptors. Fase 40 shows that the missing state is nevertheless locally causal: the mechanism does not require global information, only a short exact simulation of the radius-1 causal cone. The remaining formal problem is therefore not global dependence, but replacing this 25-cell, 12-step local computation with a symbolic account.

7.17 Minimal cone-table audit (Fase 41)

Fase 41 asks whether the successful 25-cell, 12-step cone from Section 7.16 contains a hidden smaller object: a sparse induced truth table, a reduced set of initial input bits, or a pruned causal subgraph. The analysis expands the cone into induced local updates of the form `delta_f(b,d)=f(b XOR d) XOR f(b)`, records the ordinary ECA truth-table entries used inside the cone, and computes the backwards dependency graph required to produce the final localized active defect support.

The result is deliberately negative at the table/input level. Across the 20 minimal representatives, the induced `(b,d)->d_next` tables are dense: they use 49..62 of the 64 possible local background/defect keys. All eight ordinary ECA truth-table entries are used in every representative. The final active defect support still depends on all 25 initial cone inputs. Thus there is no sparse truth-table shortcut and no input-bit elimination under this audit.

The only reduction is structural. Computing just the final active localized defect support uses 234..310 internal cone nodes, rather than all 325 nodes in the 25-cell-by-13-layer cone. This improves the circuit accounting but does not change the qualitative conclusion: the Fase 40 cone is close to minimal at the input level. The verdict is `STRUCTURAL_CONE_REDUCTION_ONLY`. The next symbolic target is therefore Boolean simplification of a dense 25-input, 12-step circuit, not further causal-support reduction.

7.18 ROBDD input-support audit (Fase 42)

Fase 42 applies reduced ordered binary decision diagrams (ROBDDs) to the dense 25-input, 12-step cone circuit. The goal is not to prove a globally minimal BDD over all possible variable orders, but to test the strongest simple Boolean reduction left open by Fase 41: whether any of the 25 cone inputs is semantically irrelevant to the active localized output functions.

For each of the 20 minimal `T=15` representatives, the Boolean circuit induced by the strict center cone is compiled into ROBDDs under the natural left-to-right cone variable order. The active localized outputs produce ROBDDs with 17,141..36,966 reachable nodes; the full 25-bit final vector produces 51,539..53,901 reachable nodes. In every representative, both the active-output functions and the full vector have support size 25/25.

The verdict is `BDD_NO_INPUT_REDUCTION`. ROBDD reduction therefore confirms the Fase 41 input-support result at the Boolean-function level: no initial cone variable can be eliminated without changing the represented output functions. BDD size remains order-dependent, so this is not a proof of global minimum BDD size over all `25!` orders. It does, however, rule out the key symbolic shortcut of Boolean input elimination. The remaining task is expression or BDD size reduction of a function that genuinely depends on all 25 inputs.

7.19 ROBDD order-sensitivity and targeted SIFT (Fase 43)

Fase 43 tests the natural follow-up to Section 7.18: if all 25 inputs are semantically necessary, perhaps the dense Boolean circuit is still compact under a better variable order. This section keeps the Fase 42 support result fixed and asks only about representation size.

Order-sensitivity preflight (Fase 43A). The three ROBDD orders already materialized in Fase 42 are compared: `natural`, `reverse`, and `center_out`. The best global order is `reverse`, but the improvement is small. Across the 20 minimal representatives, total active-output reachable nodes decrease from 552,476 under `natural` to 549,713 under `reverse`, a 0.5% reduction. The full 25-bit vector does not improve: 1,048,969 nodes under `natural` versus 1,049,085 under `reverse`. The `center_out` order is consistently poor, giving the worst active-output size in 20/20 representatives and a maximum order-sensitivity ratio of 3.119x. All tested orders preserve 25/25 active and vector support.

Targeted SIFT (Fase 43B). The most favorable representative from the preflight is `rule_73` on background `00111011`, family `F01`, with IC `00100100`. Its best Fase-42 active-output ROBDD is the `reverse` order with 16,061 reachable nodes. A one-pass variable-sifting search with checkpointing evaluates 580 orders on this representative. The best order found is `[22,21,20,19,23,24,18,17,16,15,14,13,12,11,10,9,8,7,6,5,4,3,2,1,0]`, with 16,056 active-output nodes and 52,698 full-vector nodes. This is a 5-node active-output reduction, or 0.031%, and remains far above the explicit 10,000-node compression gate. Support remains 25/25.

The verdict is not a global optimality proof over all `25!` orders, but it rules out the most direct representation shortcut tested here: simple variable reordering, even with a targeted SIFT pass on the best known candidate, does not make the dense cone compact. Any future symbolic shortcut must use a different representation or a different abstraction, not merely ROBDD variable ordering.

7.20 ANF degree audit of the T=15 cone (Fase 44)

Fase 44 tests a representation class independent from ROBDDs: algebraic normal form (ANF, or Zhegalkin polynomial) over GF(2). The question is whether the dense 25-input, 12-step causal cone has a compact polynomial representation even though BDD support and simple variable ordering do not simplify it.

The computation simulates exact truth tables in bit-packed form. Each active output has `2^25` truth values; the simulation stores those values as `uint64` blocks, then unpacks one active output at a time and applies the Mobius transform to obtain ANF coefficients. This avoids stochastic sampling and does not rely on symbolic-regression heuristics.

Across the 20 minimal `T=15` representatives, Fase 44 analyzes 174 active outputs. Active-output ANF degree ranges from 14 to 24; no output reaches the formal 25-variable ceiling. Active-output monomial counts range from 9,376 to 17,758,052. Sixty-seven of 174 active outputs have degree below 20, and every representative has at least one such lower-degree output.

The status is `LOW_OUTPUT_ANF_DEGREE_FOUND`. The cone is not algebraically uniform: some outputs remain extremely large as polynomials, but others have substantially lower degree. Fase 44 therefore turns the remaining symbolic problem from a uniform-density question into a stratification question: which outputs are algebraically simpler, and why?

7.21 ANF gradient laws (Fase 45)

Fase 45 analyzes the 174 active outputs from Fase 44 without new simulation. It finds that the ANF variation is not arbitrary; it is organized by distance from the cone center. Let `rel_pos` be the output position relative to the center cell of the 25-cell cone, and let `d = |rel_pos|`.

Degree gradient. For every active output,

`degree = 24 - d + epsilon`, with `epsilon in {0,1}`.

There are zero exceptions outside this one-bit epsilon band over all 174 active outputs. The center outputs (`d=0`) have degree exactly 24 in 13/13 cases. The immediate neighbors (`d=1`) have degree exactly 23 in 20/20 cases. The Pearson correlation between `d` and degree is -0.984525, and the linear fit is `degree ~= 24.093185 - 0.942523*d` with `R^2 = 0.969289`. The degree-24 cap also holds globally: no active output reaches degree 25.

Monomial-count gradient. Monomial counts obey an even cleaner spatial law:

`log10(monomials) ~= 7.241925 - 0.307283*d`.

The Pearson correlation between `d` and `log10(monomials)` is -0.999098, with `R^2 = 0.998197`. The slope differs from `-log10(2)` by only -0.006253, so the monomial count decays almost by a factor of two per cell away from the cone center.

The residual `epsilon` is not explained by a simple left/right symmetry. The center has `epsilon=0` in 13/13 cases; left outputs have `epsilon=1` in 32/87 cases, and right outputs in 26/74 cases. Rule identity also does not close the residual: `rule_73` has `epsilon=1` in 23/80 active outputs, while `rule_109` has `epsilon=1` in 35/94. Only 16/30 matched left/right output pairs have the same exact degree, so the defect is algebraically asymmetric even when its geometric presentation is visually paired.

The verdict is `ANF_GRADIENT_LAWS_CONFIRMED`. The `T=15` defect acts as the focus of algebraic complexity in the cone. Complexity decreases linearly in ANF degree and almost exponentially in monomial count with distance from the defect center. This structure is orthogonal to the ROBDD results of Sections 7.18-7.19: BDD support and variable ordering do not simplify the function, but ANF exposes a spatial complexity gradient invisible to those tree-based audits.

7.22 Epsilon residual audit (Fase 46)

Fase 45 leaves one open bit in the ANF degree law. The backbone `degree = 24 - |rel_pos| + epsilon` has zero exceptions, but the residual `epsilon` is not explained by sign, rule identity, or simple left/right symmetry. Fase 46 asks whether this bit has a compact predictor from static features already available in the Fase 45 records.

The audit excludes `dist=0` and `dist=1`, where `epsilon=0` in all known cases. Including those rows would inflate accuracy without explaining the residual. The remaining dataset contains 141 active outputs across the 20 minimal `T=15` representatives, with epsilon counts `{0: 83, 1: 58}` and a majority baseline of 58.87%.

Single-feature tests are evaluated by leave-one-representative-out validation. The best single feature is distance itself, at 64.89% mean accuracy. This is weak residual signal from the same variable that defines the main gradient, not a new explanatory law. The next strongest feature is `defect_phase` (64.53%), with phases 1 and 2 tending toward `epsilon=1`; this has plausible physical semantics but does not generalize strongly. `local_bg_3mer` reaches 64.18%. In contrast, `background_bit`, rule identity, and left/right sign collapse to the majority baseline, and `bg_transition` performs worse.

A depth-3 decision tree over all tested features reaches 73.05% training accuracy but only 55.65% mean leave-one-representative-out accuracy, with a 20.67% fold standard deviation. The tree's top feature is `family_id=F03`, followed by distance and local background features. This gap indicates representative/family memorization rather than a transferable epsilon rule.

The verdict is `EPSILON_REMAINS_RESIDUAL`. Fase 46 does not weaken the ANF gradient law; it separates the strong spatial backbone from a one-bit residual that is not captured by static rule, position, local-background, family, or cycle-phase descriptors. The remaining explanation, if one exists, likely requires dynamic features of the ANF computation rather than a static descriptor of the output position.

7.23 Dynamic ANF growth profile (Fase 47)

Fase 47 tests the explanation suggested by Fase 46: if `epsilon` is not predicted by static descriptors, it may be encoded in the temporal growth of the ANF itself. The computation reuses the exact bit-packed cone simulation of Fase 44, but instead of applying the Mobius transform only at the final layer, it computes ANF degree and monomial count for each active output at every layer `t = 1..12`.

The audit covers the same 20 minimal `T=15` representatives and the same 141 nontrivial residual rows with `dist >= 2`. As an internal consistency check, the recomputed `t=12` degrees and monomial counts are compared against Fase 44; there are zero mismatches.

The result is `EPSILON_DYNAMIC_RULE_FOUND`. The strongest single feature is `degree_growth_slope`, the linear slope of ANF degree over `t=1..12`. It reaches 98.58% training accuracy and 94.90% mean leave-one-representative-out accuracy with 9.44% fold standard deviation. This is a 30-point LORO improvement over the best static feature from Fase 46. `monomial_growth_slope` is weaker but still informative, with 73.37% LORO accuracy, and `t_first_full_degree` reaches 71.96%.

A depth-3 decision tree over the dynamic features reaches 86.52% training accuracy and 85.01% mean leave-one-representative-out accuracy. Its first split is `t_first_full_degree <= 11.5`, which predicts `epsilon=0`; outputs that do not reach their final degree before the last cone layer require additional monomial-growth and left/right-slope information. This gives a physical interpretation of the residual: `epsilon=1` is associated with outputs whose algebraic degree remains active near the causal horizon.

This is not a static pre-computation shortcut. The strongest feature uses the complete temporal degree trajectory through the final cone layer. The result is therefore best interpreted as a dynamic full-profile law of ANF growth: the epsilon bit is not visible in static output descriptors, but it is strongly encoded in how algebraic degree accumulates across the 12-step cone.

7.24 Early dynamic ANF horizon audit (Fase 48)

Fase 48 asks whether the Fase 47 dynamic predictor can be made earlier. The test reuses the stored `t=1..12` ANF histories from Fase 47; no new ECA or cone simulation is performed. For each horizon `K in {6,8,9,10,11,12}`, it recomputes the future-blind feature `degree_growth_slope_K` using only the degree profile from `t=1..K`, and evaluates it with the same 20-fold leave-one-representative-out protocol over the 141 nontrivial residual rows with `dist >= 2`.

The result is `FULL_PROFILE_REQUIRED`. The horizon table for `degree_growth_slope_K` is:

Horizon K	LORO accuracy
6	61.74%
8	76.56%
9	75.27%
10	76.09%
11	79.47%
12	94.90%

The transition from `K=11` to `K=12` is not gradual: the final cone layer adds about 15 points of predictive accuracy. This supports the interpretation that `epsilon` is decided at the causal horizon. Outputs with `epsilon=1` are not merely high-complexity outputs; they are outputs whose ANF degree trajectory still carries decisive information in the last step of the 12-step cone.

The audit also tracks `t_first_full_degree_K`, but that feature uses the final expected degree from Fase 44 and is therefore not fully future-blind. The clean feature is `degree_growth_slope_K`. Under that feature class, no horizon before `K=12` reaches the 90% gate. Thus Fase 47's law is not an early dynamic shortcut; it is a full-horizon profile law.

7.25 Generalization of the ANF gradient law (Fase 49)

Fase 49 tests whether the ANF gradient law discovered in Sections 7.20-7.21 is specific to the original length-8 `T=15` representatives, or whether it also holds for the external `T=15` backgrounds found in Fase 34. The external set contains eight replay-verified backgrounds: one length-9 `rule_73` background, five length-10 `rule_73` backgrounds, and two length-10 `rule_109` backgrounds. These are genuine external backgrounds, not rotations of the length-8 set.

The protocol reuses the exact bit-sliced Mobius ANF machinery from Fase 44: `WINDOW_CELLS=25` and `T_WINDOW=12`. Before ANF evaluation, each external background is replay-verified as `T=15` using its minimal witness IC from Fase 34. All 8/8 backgrounds pass this replay gate. The Fase 34 witnesses have varying active defect widths, so the test does not assume constant visual defect width; the 25-cell cone is fixed by the radius-1, 12-step causal horizon.

Across the eight external backgrounds, Fase 49 analyzes 63 active outputs. The degree band remains exact:

`degree = 24 - d + epsilon`, with `epsilon in {0,1}`.

There are 0/63 exceptions. The active degree range is 16..24, and epsilon counts are `{0: 39, 1: 24}`.

The monomial-count law also generalizes quantitatively. The external fit is

`log10(monomials) ~= 7.224069 - 0.302890*d`, with `R^2 = 0.998263`.

Compared with the length-8 reference `log10(monomials) ~= 7.241925 - 0.307283*d`, the intercept differs by 0.25% and the slope magnitude by 1.43%. This is well inside the predefined generalization gate.

The verdict is `ANF_GRADIENT_GENERALIZES`. The ANF gradient is therefore not a length-8-specific artifact, at least across the original length-8 representatives and the external length-9/10 witnesses tested here. The question of whether this gradient is specific to the `T=15` period or to a broader `rule_73`/`rule_109` mechanism family is addressed in Section 7.26.

7.26 Specificity of the ANF gradient beyond T=15 (Fases 50–53)

Fases 50--53 test whether the ANF gradient law from Fases 44--49 is a generic consequence of period, cone size, or active-support width, or whether it is tied to a more specific mechanism family.

Compact T=2 baselines (Fases 50--51). Fase 50 first tests the cleanest stationary local period-2 oscillator known from the quiescent-background catalog: `rule_108` on a zero background. Its active footprint is compact (`#.# <-> ###`), and at the 12-step comparison horizon the concrete active support has only 2 outputs in a single distance class. This is not enough support for a spatial active-output gradient. Fase 51 repeats the test for the four right-moving T=2 glider representatives `rule_20`, `rule_52`, `rule_148`, and `rule_180`, using the exact catalog ICs and a comoving final frame. The one-cell phase produces one active output and the two-cell phase control produces two adjacent active outputs, again in a single comoving distance class. The verdict for these compact T=2 baselines is `ANF_GRADIENT_T15_SPECIFIC`, but for a precise reason: compact active support prevents a meaningful gradient test. This does not by itself prove that the gradient is tied to period 15.

Wide periodic-background non-T15 cases (Fase 52). Fase 52 therefore moves to wide stationary oscillators over nontrivial periodic backgrounds. These cases have multiple active distance classes and are measured as XOR defects relative to the background orbit, matching the T=15 convention. The key positive witness is `rule_109` on background `1011`, with `T_local=10`. At the common 12-step horizon it gives

`log10(monomials) ~= 7.258688 - 0.307674*d`, with `R^2 = 0.999349`.

The slope differs from the T=15 reference `-0.307283` by only 0.13%. This is not merely a similar trend; within this finite audit it is the same spatial monomial-decay gradient. Other non-T15 periodic-background cases are weaker or ambiguous: `rule_73` on background `0010` with `T_local=10` has too few active distance classes at the 12-step horizon (`reliable=no`); `rule_73` on background `0011` with `T_local=12` gives slope `-0.296180` but only `R^2 = 0.877281`; and `rule_94` on background `0010` with `T_local=3` is flat at its own period (slope `0.000523`, `R^2 = 0.049180`) but develops a slope `-0.341994` with `R^2 = 0.955626` only when oversampled to `T_WINDOW=12`. That last case is treated as a horizon effect rather than the same natural period law. The Fase 52 verdict is `ANF_GRADIENT_MECHANISM_DEPENDENT`.

External-family test (Fase 53). Fase 53 then tests wide candidates outside the `rule_73`/`rule_109` family, using the shortest catalog witnesses at maximum support `span=11`. `rule_54` on background `0010` with `T_local=4` has slope `-0.002354` at its own period: effectively flat. `rule_94` on background `0001` with `T_local=6` has slope `0.000000` and `R^2 = 0.000000` at its own period. `rule_133` on background `1011` with `T_local=6` also has slope `0.000000` and `R^2 = 0.000000` at its own period. These own-period tests have enough active support to be informative; the flat result is not a support artifact. At the common 12-step horizon, some external cases acquire nonzero slopes, but they do not reproduce the T=15-quality combination of support reliability, slope, and `R^2`. The Fase 53 verdict is `ANF_GRADIENT_FAMILY_73_109`.

Interpretation. The ANF gradient is not a consequence of period, cone size, or active-support width alone. It is mechanism-dependent. The confirmed non-T15 witness in this audit is the `rule_109`/background `1011`/`T=10` case, while compact T=2 oscillators and the external families tested (`rule_54`, `rule_94`, and `rule_133`) do not reproduce the T15-quality gradient at their natural periods. Section

7.27 tests whether this non-T15 witness broadens across additional `rule_73`/`rule_109` family members. No claim is made about untested ECA families.

7.27 Additional family robustness test for the ANF gradient (Fase 54)

Fase 54 tests whether the non-T15 gradient witness from Section 7.26 is an isolated case or part of a broader natural-period law within the `rule_73`/`rule_109` periodic-background family. The test uses the same 25-input bit-sliced Mobius ANF engine as Fases 52--53. The primary criterion is the natural-period horizon `T_WINDOW=T_local`; the common 12-step horizon is reported only as a secondary comparison.

Three additional family witnesses are selected from the periodic-background catalog:

- `rule_109`, background `1011`, `T_local=6`, IC `00001001`;
- `rule_109`, background `1101`, `T_local=10`, IC `0001000`;
- `rule_73`, background `0010`, `T_local=6`, IC `1100111`.

At their natural periods, none reproduces the T15-quality gradient. The two `T=6` cases are essentially flat: `rule_109`/`1011` has slope `0.000026` and `R^2 = 0.604938`, while `rule_73`/`0010` has slope `-0.000027` and `R^2 = 0.604938`. The `rule_109`/`1101`/`T=10` case has a nonzero slope (`-0.209698`) but lower fit quality (`R^2 = 0.880488`) and a 31.76% deviation from the T15 reference slope, so it is not comparable to the T15 law.

At the common 12-step horizon, two of the new cases do show T15-like slopes: `rule_109`/`1011`/`T=6` gives slope `-0.303174` with `R^2 = 0.999487`, and `rule_73`/`0010`/`T=6` gives slope `-0.320463` with `R^2 = 0.999687`. Because these gradients appear only after oversampling to the 12-step horizon, they are classified as horizon effects rather than natural-period witnesses. The Fase 54 verdict is therefore `ANF_GRADIENT_ISOLATED_WITNESS` under this targeted three-case test: the strong non-T15 natural-period witness remains the `rule_109`/background `1011`/`T=10` case from Fase 52. Section 7.28 then replaces this local conclusion with a full catalog census.

7.28 ANF gradient census across the periodic-background catalog (Fase 55)

Fase 55 turns the targeted robustness question into a catalog-level census. The test scans all stationary periodic-background oscillator groups with `span >= 11`, excluding compact `T_local=2` baselines and the original `T_local=15` family. One IC is evaluated per `(rule, background, T_local)` group: groups already tested in Fases 52--54 keep their exact previous ICs as consistency baselines, while new groups use maximum span and then shortest word as the tie-breaker.

The preflight census contains 66 groups across six rules:

- period distribution `{3: 18, 4: 8, 6: 22, 8: 6, 10: 8, 12: 4}`;
- rule distribution `{54: 4, 73: 17, 94: 12, 109: 17, 133: 12, 147: 4}`;
- seven groups marked `already_tested=true`.

The ANF engine performs 128 measurements, with zero packed/concrete discrepancies. The category counts are:

Category	Count
`NATURAL_PERIOD_STRONG`	2
`HORIZON_ACCEPTABLE`	3
`HORIZON_ARTIFACT`	20
`INSUFFICIENT_SUPPORT`	3
`NEGATIVE`	38

Among previously untested cases, the census finds two `NATURAL_PERIOD_STRONG` witnesses, both in `rule_109`:

Case	Slope	R^2	Delta vs T15
`rule_109`/background `0011`/`T=12`/IC `10010100`	-0.298274	0.998341	2.93%
`rule_109`/background `1100`/`T=12`/IC `00101001`	-0.298274	0.998341	2.93%

Because these cases have `T_local=12`, their natural-period measurement and the common `T_WINDOW=12` measurement are the same run. They are valid natural-period witnesses, but they are not two independent confirmations across different horizons.

The census also finds three `HORIZON_ACCEPTABLE` cases: the already-tested baseline `rule_109`/background `1011`/`T=10` and two new `rule_109`/`T=8` witnesses:

Case	Natural-period fit	Common-horizon fit
`rule_109`/background `0110`/`T=8`/IC `0000011`	slope `-0.106802`, R^2 `0.617294`	slope `-0.298928`, R^2 `0.998276`, delta `2.72%`
`rule_109`/background `1100`/`T=8`/IC `00000110`	slope `-0.106802`, R^2 `0.617294`	slope `-0.298928`, R^2 `0.998276`, delta `2.72%`
`rule_109`/background `1011`/`T=10`/IC `00000001`	slope `-0.196127`, R^2 `0.922575`	slope `-0.307674`, R^2 `0.999349`, delta `0.13%`

The negative pattern is equally important. No `rule_73` case reaches `NATURAL_PERIOD_STRONG` or `HORIZON_ACCEPTABLE`. The external rules tested (`rule_54`, `rule_94`, `rule_133`, and `rule_147`) also produce no strong or acceptable witness. Many `T_local <= 6` cases are `HORIZON_ARTIFACT`: they develop T15-like slopes only after oversampling to the 12-step horizon, not at their natural period.

The Fase 55 verdict is `NEW_NATURAL_PERIOD_WITNESS_FOUND`. This updates the interpretation from Sections 7.26--7.27: the ANF gradient is not merely an isolated non-T15 witness, but neither is it shared symmetrically by `rule_73`/`rule_109` as a family. In the censused catalog, the robust non-T15 evidence is concentrated in `rule_109`.

8. Observer Artifacts and Pipeline Equivariance

The ZUSE pipeline contains two classes of observer artifact that the atlas identifies and characterizes. Framing these artifacts as results — not just implementation limitations — is important: they define the boundary between what the system measures reliably and what it does not.

8.1 Mirror asymmetry in `tipo_unico` (Fase 6b)

`rule_110` and `rule_124` are left-right mirrors of each other: the rule table for `rule_124` is obtained by reflecting every neighborhood `f(l,c,r) -> f(r,c,l)` in the `rule_110` table. Under periodic boundary conditions, the two rules produce physically equivalent dynamics up to spatial reflection.

Despite this equivalence, `tipo_unico` can fire asymmetrically: it may be accepted for one orientation and rejected for the other, depending on which structure types the heuristic observers label from the specific frame sequence. Since `tipo_unico` counts whether exactly one structure type appears, its value depends on the labeling convention of the observers, not only on the CA dynamics.

`tipo_unico` is retained in the atlas for its exploratory value: it reliably distinguishes runs with homogeneous structure populations from runs with mixed populations. It should not be used as evidence of physical left-right asymmetry.

8.2 Translation non-equivariance of the dedup pipeline (Fase 19)

ECA with periodic boundary conditions is translation-invariant: shifting the initial condition by any number of cells produces the same dynamics up to a spatial shift. A pipeline that correctly identifies physical structures should therefore return the same structure count for all translations of the same IC.

Fase 19 tested this directly: 64 single-bit ICs for `rule_54`, one active bit at each position `k = 0..63`, with `width = 64` and `steps = 96`. The ECA frames were confirmed translation-invariant (frame identity after shift normalization: `True`). The observer/dedup pipeline was not:

metric	range across 64 positions
`dedup_structure_count`	`15..24` (29 distinct result classes)
`raw_structure_count`	`45..72`
law signature	`temporal_scale_stability` (all 64 identical)
`analysis_status`	`ok` (all 64)

metric	value
ICs tested	64 single-bit positions (`k = 0..63`)
ECA translation-invariant	True
Unique result classes (observer)	29
dedup range observed	15-24
Noise gate threshold	> 40
Closest single-bit IC to gate	`k = 55`, `dedup = 24`
Complex-IC reference (Fase 13)	`dedup = 32..39`
Law signature (all 64 positions)	`temporal_scale_stability`
Conclusion	noise-gate crossing requires complex IC geometry; single-bit ICs stay at least 16 below gate

The mechanism is a boundary interaction: `rule_54` produces wide-spreading patterns that cross the periodic frame boundary. The dedup algorithm's handling of cyclic-span structures varies depending on the absolute IC position relative to where structure boundaries fall on the lattice. The result is a position-dependent count that is not a translation-equivariant physical observable.

Fase 25 isolates a second instance of the same geometric failure mode in the local-oscillator detector. Co-translated background and XOR perturbation orbits are identical in 80/80 runs, but `linear_shape` loses 22 moving signatures when the localized difference straddles positions 255 and 0. Circular canonicalization restores 80/80 signatures. This directly attributes the non-equivariance to linear treatment of a periodic lattice.

8.3 Implications for the atlas

Both artifacts are bounded in their effect:

- `tipo_unico` asymmetry is a labeling artifact, not a count artifact. It affects which laws are accepted, but only for runs where the structure population is near the one-type boundary. Runs with clearly homogeneous or clearly mixed populations are unaffected.
- Dedup non-equivariance affects absolute structure counts but not law signatures. In Fase 19, all 64 translated ICs produce identical law signatures despite varying dedup counts. The noise gate (`dedup > 40`) is never approached by single-bit ICs, and law evaluation depends on count magnitude only through the gate.

The atlas therefore relies on law signatures as the primary evidence unit. Absolute dedup counts appear in world profiles as context and should be interpreted with the translation-equivariance caveat. Future work on symmetry-invariant observers would remove both artifacts.

9. Limitations

9.1 Fixed protocol parameters

The atlas is valid for the parameter regime used: `width = 64` (formal profiles), `width = 128` (rule_108 oscillator), `steps` roughly `24..200`, and the IC protocols defined per world. The two calibrated thresholds — `frontera_temporal` upper bound `0.4352` and `temporal_scale_stability` threshold `19.03` — were fit on data from this regime. Applying the atlas to significantly different widths or step counts requires recalibration. This is not a flaw in the methodology; it is the expected scope of an empirically grounded atlas.

9.2 Heuristic observers

The observer stack uses geometric heuristics to label structures as `glider`, `bloque`, or `oscilador`. These heuristics are not derived from first principles and are not provably complete or sound for arbitrary ECA dynamics. As shown in Section 8, they are not translation-equivariant for wide-spreading patterns and are not mirror-invariant for `tipo_unico`. The atlas is built on law signatures, which are more robust than absolute observer counts, but the underlying observers remain heuristic. Replacing them with symmetry-invariant observers would be a meaningful improvement.

9.3 Bounded local oscillator protocol

The uniqueness claim for `rule_108` holds under a specific protocol: quiescent zero background, stationary exact periodicity (no drift), IC words of binary length 1..12 (502 words of length 1..8 plus 7,676 words of length 9..12), and period detection window 2..16 with local span <= 32. IC words longer than 12 or longer detection periods are outside the current stationary protocol. A separate periodic-background sweep (Section 7.6) confirms that non-zero backgrounds substantially change the oscillator landscape; those results define a different regime and do not apply to the zero-background uniqueness claim. Moving oscillators under quiescent zero background were searched in a companion sweep -- Section 7.5 reports the result. The claim is therefore: no other quiescent ECA rule produces a stationary local-period oscillator under this protocol. It is not a claim about ECA oscillators in general.

The period-8 background sweep (Section 7.7) is also bounded by background phase. Fase 25 completed the strict co-translation test: the physical orbits are exact translations in 80/80 runs, while the original linear observer recovers 58/80 signatures because of cyclic boundary crossings. Circular canonicalization recovers 80/80. Fixed-IC phase dependence remains in all 10 sampled cases after correction, so it is physical alignment sensitivity rather than pure observer non-equivariance.

Fase 26 strengthens the longest-period result without converting it into a general theorem. All 20 minimal `T=15` rule/background representatives persist through step 900 and share a background temporal period of 3, but only 23/160 background phases and 4/134 one-bit IC mutations retain `T=15`. The result is therefore robust in time and narrow in basin under the tested protocol.

Fase 27 establishes the state-cycle mechanism of the 5:1 locking computationally. After burn-in, the localized defect `D(t) = X(t) XOR B(t)` cycles through exactly five distinct states under `F^3` across all 20 minimal representatives (Section 7.9). The 5:1 ratio is therefore the cycle length of the defect under the three-step operator, not a spatial resonance. The symbolic derivation of why this five-cycle appears in `rule_73/rule_109` but not in other rules over `T_bg=3` backgrounds remains open.

Fase 28 proves the black/white conjugation of the defect dynamics analytically and verifies it exhaustively at local and orbit levels (Section 7.10). It also rules out one tempting simplification: no fixed sparse subset of induced local transitions is shared by all 100 `F^3` edges. The remaining symbolic problem requires phase-aware or higher-order block variables.

Fase 29 and Fase 30 sharpen that boundary. The T=15 cycle is not a pure defect-only dynamic: canonical defect shapes are background-specific already at `W=0`, and no nontrivial fixed local block signature is shared across all backgrounds. But the variation is not arbitrary. The 20 representatives collapse into 13 phase-rotated defect-cycle families, including two families shared across `rule_73` and `rule_109`. The open problem is therefore not simply "find the local rule"; it is to map the temporal background orbit and IC alignment to a finite shape family and phase offset.

Fase 31 and Fase 32 provide the current compact description of that map. No tested background-only descriptor shorter than the full temporal orbit predicts the 13 families globally. However, conditioned on rule identity, the circular multiset of length-4 background subwords separates the families, and rotational variants preserve the predicted family in 140/140 co-translated runs. The same rotations fail almost completely with fixed IC placement (3/140 detections), so IC/background alignment is an essential physical variable, not a bookkeeping detail. Fase 33 then shows that length 8 contains no unseen same-descriptor backgrounds outside the known representatives and their rotations; the two length-4 descriptor collisions that do exist are already family-preserving. Fase 34 moves to primitive length-9/10 backgrounds with `T_bg=3` and finds 90 T=15 detections across 8 external backgrounds. This confirms that the five-to-one mechanism generalizes beyond length 8, while leaving the variable length descriptor problem open.

Fases 35--38 replace visual shape families with explicit macro-transition tables and effective orbit embeddings. The induced `F^3` transition table is a sufficient discriminator and refines the 13 visual families. The `F00` table identity is explained by convergence to the same effective period-3 background orbit, but that explanation is too coarse globally: each rule has only one canonical period-3 orbit across the confirmed set. The first sufficient post-burn-in descriptor is therefore `(rule, sample_orbit_step, sample_rotation_offset, defect_state0)`, where `defect_state0` is the first sampled stable-cycle defect state.

Fase 39 tests whether that final measured variable can be predicted compactly from pre-burn-in data. The answer is negative under the tested descriptors. All 20 representatives enter the stable five-cycle by `t=12` and 15/20 enter at `t=3`, but exact pre-burn-in predictors either rely on post-hoc entry time or collapse into singleton case identifiers. The remaining formal problem is therefore precise: predict `defect_state0` from the raw background/IC pair without effectively replaying the early transient.

Fase 40 shows that "effectively replaying" does not require the full system. A strict 25-cell causal cone centered on the IC and simulated for 12 steps matches the full defect state at `t=12` in 20/20 representatives and, after phase projection, recovers `defect_state0` at `t=81` in 20/20. The compression relative to the full 256-by-81 simulation is 69.1x. Thus the unresolved pre-burn-in component is locally causal rather than global: the open question is how to replace the 25-cell, 12-step cone computation with a symbolic rule.

Fase 41 audits whether that local computation contains an obvious smaller support. It does not. All 25 initial cone inputs are required for the final active defect support, all eight ordinary ECA table entries are used in every representative, and the induced background/defect tables are dense (49..62 of 64 possible keys). The cone has only structural pruning for active outputs (234..310 internal nodes instead of 325), so the remaining symbolic problem is Boolean minimization of a dense local circuit, not discovery of a sparse subtable or smaller causal support.

Fase 42 verifies the input-support conclusion at the Boolean-function level. Natural-order ROBDDs for the active localized output functions contain all 25 support variables in every representative, with 17,141..36,966 reachable nodes for active outputs and 51,539..53,901 for the full 25-bit vector. Variable support is semantic, not merely an artifact of the natural BDD order, so this rules out Boolean input elimination. It does not claim a globally minimal BDD size over all variable orders.

Fase 43 then tests the most direct BDD-size escape route. Existing `natural`, `reverse`, and `center_out` orders show only weak useful sensitivity: `reverse` improves total active-output nodes by 0.5%, while `center_out` is worst in 20/20 representatives. A checkpointed one-pass SIFT search on the most favorable representative evaluates 580 orders and improves 16,061 active-output nodes to 16,056, far above the 10,000-node compression gate. This does not prove global BDD-size optimality, but it rules out simple variable-ordering as the missing symbolic shortcut.

Fases 44-45 then switch representation class from BDDs to ANF polynomials. They show that the Boolean cone is not algebraically uniform. Active-output degree obeys `degree = 24 - |rel_pos| + epsilon`, with `epsilon in {0,1}` and zero exceptions over 174 outputs; monomial counts decay with `log10(monomials) ~= 7.241925 - 0.307283*|rel_pos|` (`R^2=0.998197`). This spatial gradient is centered on the defect and is orthogonal to the BDD negative results. It opens a sharper symbolic question: whether the gradient can be derived from cone geometry and the background orbit, and what determines the residual epsilon for distances >= 2.

Fase 46 tests that residual directly. After removing `dist=0` and `dist=1`, where `epsilon=0` trivially, it evaluates 141 active outputs using static features from rule identity, signed position, local background bits, family id, and defect phase. The best single feature reaches only 64.89% leave-one-representative-out accuracy, and a depth-3 decision tree falls from 73.05% training accuracy to 55.65% leave-one-representative-out accuracy. Thus the epsilon bit remains residual under the current static feature class.

Fase 47 then changes the feature class from static descriptors to dynamic ANF growth profiles. Recomputing ANF degree and monomial count at each cone layer `t=1..12` yields zero mismatches against the final Fase 44 degrees at `t=12`. The single feature `degree_growth_slope` predicts `epsilon` with 94.90% leave-one-representative-out accuracy. The result should not be read as a pre-computation shortcut, because the feature uses the full temporal trajectory through the final layer; it is instead a dynamic law explaining where the one-bit residual lives.

Fase 48 then asks whether that dynamic law appears before the final cone layer. The answer is negative under the tested horizon protocol: the future-blind `degree_growth_slope_K` feature reaches 79.47% LORO accuracy at `K=11`, then jumps to 94.90% at `K=12`. Thus the epsilon bit is a full-horizon ANF-growth property rather than an early dynamic shortcut.

9.4 Empirical atlas, not axiomatic classification

The world categories are induced from observed law signatures across a finite number of seeds and step counts. A world classified as `multiregimen-productivo` on 6..15 visits could exhibit different behavior at larger scale, with different IC distributions, or under longer runs. The categories are stable empirical summaries, not theorems. `rule_90` is a clear example: it is classified as `multiregimen-escala-dependiente` because high-scale visits become silent under the current protocol, but the underlying XOR dynamics have algebraic structure that the current seven laws do not capture.

Fase 20 gives the same warning for `frontera_temporal`: 24 additional rules were rich in `frontera_temporal` at sweep scale, but the top four failed long-journal validation. Category assignment is therefore protocol-scale dependent; short-scale richness is candidate evidence, not atlas-grade classification.

Fase 21 gives an analogous warning for `periodicidad`. With random ICs, production `periodicidad` is rare in ECA; with explicitly periodic IC families, it appears in `207/256` rules. The law is therefore not inaccessible to ECA, but it is strongly conditioned by the IC family. Atlas rows should be read as claims about a stated rule/IC/scale protocol, not as intrinsic properties of the rule table alone.

9.5 PySR symbolic regression -- negative result

The decision-tree analyses (Section 4, temporal calibration) provide strong empirical signal but not closed-form symbolic expressions. PySR was planned as a follow-up to produce interpretable formulas for the calibrated thresholds and fragility spectra. PySR 1.5.10 was subsequently unblocked. A full regression over 15 atlas worlds (five features: `mean_laws`, `peak_diversity`, `noise_ratio`, `non_empty_ratio`, `f_core`; target `f_total`; 40 iterations) produced a best expression of complexity 9 with `MSE = 0.035`, above the threshold for a paper-worthy finding. The dominant predictor is `f_core`; the residual is driven by `rule_108` as a structural outlier (`f_gap = 0.945`) that is not predictable from aggregate features without a mechanism feature. The symbolic layer remains incomplete, and this negative result is consistent with the `f_core`/`f_gap` separation documented in Section 6.

10. Next Work

10.1 Symmetry-invariant observers

The two observer artifacts identified in Section 8 — `tipo_unico` mirror asymmetry and dedup translation non-equivariance — share a root cause: the heuristic observers do not encode the symmetries of the underlying CA. A natural next step is to build observers that canonicalize structure representations under spatial reflection and translation before counting. This would make `tipo_unico` a mirror-invariant physical property and dedup counts stable across IC positions, strengthening the evidential basis for both the atlas and the fragility measurements.

10.2 Extended local oscillator search

The `rule_108` uniqueness result holds under the current stationary protocol (zero background, exact period, IC words of length <= 12, span <= 32). Several controlled extensions have now been completed:

- **Moving oscillators**: completed. A companion sweep over all 128 quiescent rules found eight rules producing minimal period-2 speed-1 gliders (Section 7.5). No longer-period or slower-speed moving oscillators were found under this protocol. The follow-up length-9..12 sweep found no additional stationary or moving oscillator rules.
- **Longer IC words beyond 12**: extend the IC sweep from length 12 to length 16 or beyond to test whether substantially wider seed patterns produce oscillators in rules that failed the length-12 protocol.
- **Non-zero backgrounds**: completed. A sweep over 15 non-zero periodic backgrounds (template lengths 1, 2, 4) across all 256 rules and 502 IC words (1,927,680 runs) found that the periodic-background regime is substantially richer than the quiescent regime: 30 stationary rules, 36 moving rules, including period-4 oscillators and speed-0.5 gliders not present under zero background. `rule_108` persists under all-one background with the same motif. Full results in Section 7.6.
- **Period-8 backgrounds**: completed. A sweep over 30 primitive length-8 binary necklaces (3,855,360 runs) found 4 new stationary rules, 19 new moving rules, five new period classes (T=6, 8, 10, 12, 15), and a new glider speed (2/3 cell/step, T=3). All 10 sampled results were background-phase dependent after circular-geometry correction. Fase 25 confirms exact co-translation equivariance in 80/80 runs with circular shape canonicalization. Full results are in Section 7.7.
- **T=15 anatomy**: completed. All 221 detections belong to the reflection-symmetric, black/white-conjugate pair `rule_73/rule_109`. The 20 minimal rule/background representatives remain exact through step 900 and lock at five times the background temporal period (`15/3`). The basin is narrow: 23/160 background phases and 4/134 one-bit IC mutations retain `T=15`. Full results are in Section 7.8.
- **Five-state locking mechanism**: completed. The localized defect cycles through five distinct states under `F^3`. The ratio `T_local/T_bg=5` is the defect cycle length, confirmed in all 20 minimal representatives. Full results are in Section 7.9.
- **Induced defect algebra**: partial. The exact law `delta_f(b,d)=f(b XOR d) XOR f(b)` proves that the `rule_73` and `rule_109` defect orbits are identical under simultaneous black/white conjugation. Profiling all 100 `F^3` edges rejects a universal sparse-entry explanation and points instead to spatial phase or higher-order block states. Full results are in Section 7.10.

- **Block locality and shape families**: partial-positive. Fixed local defect blocks do not explain the T=15 cycle: the active defect already carries background-specific context at `W=0`, and no nontrivial shared block signature exists across all backgrounds. The variation is nevertheless finite: 20 representatives collapse into 13 phase-rotated shape families, with two families shared across the conjugate rules. Full results are in Section 7.11.
- **Compact T=15 state variable**: completed for the known representative set. No compact background-only descriptor predicts all 13 families globally, but the rule-conditioned descriptor `subpatterns_len4` separates all backgrounds per rule. Rotation generalization validates the compact state variable `(rule, subpatterns_len4, IC/background alignment)`: 140/140 co-translated rotations preserve both T=15 and the predicted family, while fixed-IC rotations collapse to 3/140 detections. Full results are in Section 7.12.
- **External T=15 validation**: completed for targeted primitive length-9/10 backgrounds. Fase 33 proves that length 8 has no unseen same-descriptor backgrounds outside rotations of the known set. Fase 34 then targets the 66 primitive length-9/10 backgrounds with `T_bg=3` under `rule_73/rule_109` and finds 90 T=15 detections across 8 external backgrounds. Full results are in Section 7.13.
- **Transition-table and embedding analysis**: completed for the 20 minimal `T=15` representatives. The explicit `F^3` transition table is a sufficient discriminator and refines the visual family partition. Family `F00` is explained by convergence to a shared effective period-3 background orbit, but canonical orbit identity alone is too coarse globally. The first sufficient post-burn-in descriptor is `(rule, sample_orbit_step, sample_rotation_offset, defect_state0)`. Full results are in Section 7.14.
- **Pre-burn-in entry phase**: negative delimiter. The defect enters the stable five-cycle quickly (all 20 by `t=12`, 15/20 at `t=3`), but the tested pre-burn-in descriptors do not give a compact predictor of the entry phase: exact descriptors are mostly singleton case identifiers or post-hoc measurements. Full results are in Section 7.15.
- **Early causal-cone predictor**: positive. The strict `2t+1` causal cone centered on the IC at `t=12` (25 cells for 12 steps) matches the full defect state in 20/20 representatives and recovers `defect_state0` after phase projection in 20/20. This is a 69.1x compression relative to the full 256-by-81 simulation. Full results are in Section 7.16.
- **Minimal cone-table audit**: negative/structural. The Fase 40 cone has no sparse table shortcut: induced `(b,d)->d_next` tables use 49..62 of 64 keys, all eight ordinary ECA entries are used, and all 25 initial cone inputs are required for the active localized output. The only reduction is structural: active-output dependency uses 234..310 internal nodes instead of the full 325. Full results are in Section 7.17.
- **ROBDD cone audit**: negative/semantic. Reduced ordered BDDs under the natural cone order confirm that the active-output functions depend on all 25 inputs in every representative. Active-output BDDs have 17,141..36,966 reachable nodes; the full 25-bit vector has 51,539..53,901. This rules out Boolean input elimination, while not claiming globally minimal BDD size over all variable orders. Full results are in Section 7.18.
- **ROBDD order search**: negative/bounded. The `reverse` order improves total active-output nodes by only 0.5% over `natural`, while `center_out` is worst in 20/20 representatives. A targeted 580-order SIFT pass on the best known representative improves 16,061 active nodes to 16,056 (0.031%), missing the 10,000-node compression gate. Full results are in Section 7.19.
- **ANF gradient**: positive/structural. Exact ANF analysis of 174 active outputs shows that `degree = 24 - |rel_pos| + epsilon`, with `epsilon in {0,1}` and zero exceptions, while monomial counts decay almost by a factor of two per cell from the cone center (`R^2=0.998197`). The epsilon residual is tested directly in Fase 46: after excluding trivial `dist=0,1` rows, the best single-feature predictor reaches 64.89% leave-one-representative-out accuracy, and a depth-3 tree reaches only 55.65% under the same validation. The residual remains unexplained by static rule/position/background/family features. Fase 47 resolves the residual at the dynamic-profile level: `degree_growth_slope` over `t=1..12` predicts epsilon with 94.90% leave-one-representative-out accuracy, with zero mismatches against Fase 44 at `t=12`. Fase 48 shows that this is a full-horizon effect: `K=11` reaches only 79.47% LORO accuracy, while `K=12` jumps to 94.90%. Fase 49 validates the gradient on external length-9/10 `T=15` witnesses (0/63 exceptions, `R^2=0.998263`). Fases 50--53 then test specificity outside the original T=15 set: compact T=2 baselines lack enough active support; `rule_109`/`T=10` reproduces the T=15 slope within 0.13%; and the external `rule_54`, `rule_94`, and `rule_133` families are flat at their natural periods. Fase 54 then tests additional `rule_73`/`rule_109` family witnesses and temporarily isolates the known `rule_109`/background `1011`/`T=10` case. Fase 55 replaces that local conclusion with a full catalog census: 66 non-T15 groups with `span >= 11`, 128 ANF measurements, two new `rule_109`/`T=12` natural-period witnesses, and two new `rule_109`/`T=8` acceptable-horizon witnesses. No `rule_73` or external-rule case becomes a strong/acceptable witness. Full results are in Sections 7.20-7.28.

Each extension is a controlled experiment with the same measurement protocol; only the IC or background definition changes.

10.3 PySR symbolic regression

The decision-tree calibration for `frontera_temporal` and `temporal_scale_stability` (Section 4) provides thresholds but not formulas. PySR symbolic regression on the fragility spectrum (`f_total`, `f_core`, `f_gap` as functions of rule properties and IC metrics) is now technically available, but the first atlas-wide run did not produce a compact paper-worthy formula. The next useful symbolic-regression target would require mechanism features rather than aggregate atlas features alone.

10.4 Figures

The following six figures are planned for the preprint draft:

1. **World taxonomy table** — the full 20-world atlas with categories, law coverage symbols, and fragility columns, formatted as a paper-ready table.

2. **Law coverage matrix** — the `✓ / · / - / ?` matrix from `outputs/world_taxonomy/law_map.md`, rendered as a heatmap or binary grid.
3. **`f_total` / `f_core` spectrum** — a two-axis scatter or bar chart showing all measured worlds positioned by `f_total` and `f_core`, with the four fragility mechanisms labeled.
4. **`rule_108` oscillator motif** — a space-time diagram of the `#.# <-> ###` two-step cycle, showing several periods on a quiescent background.
5. **`rule_54` gate and observer non-equivariance** — a dual figure: the Fase 13 noise-gate crossing diagram (reference dedup vs perturbed dedup) alongside the Fase 19 per-position dedup variation (15..24 across k=0..63).
6. **Moving oscillator space-time diagram** -- dual panel showing `rule_20` (right-moving) and `rule_6` (left-moving) over 8 time steps, with active cells dark and quiescent background light. Illustrates the `[0] <-> [0,1]` glider cycle and the `+/-2` drift per period.

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